CHAPTER 3

Inference for Quantile Regression

This chapter provides a practical guide to statistical inference for quantile regression applications. In the earlier chapters, we have described a number of applications of quantile regression and provided various representations of the precision of these estimates; in this chapter and the one to follow, we will describe a variety of inference methods more explicitly. There are several competing approaches to inference in the literature and some guidance will be offered on their advantages and disadvantages. Ideally, of course, we would aspire to provide a finite-sample apparatus for statistical inference about quantile regression like the elegant classical theory of least-squares inference under independently and identically distributed (iid) Gaussian errors. But we must recognize that even in the least-squares theory it is necessary to resort to asymptotic approximations as soon as we depart significantly from idealized Gaussian conditions.

Nevertheless, we will begin by briefly describing what is known about the finite-sample theory of the quantile regression estimator and its connection to the classical theory of inference for the univariate quantiles. The asymptotic theory of inference is introduced with a heuristic discussion of the asymptotic behavior of the ordinary sample quantile; then a brief overview of quantile regression asymptotics is given. A more detailed treatment of the asymptotic theory of quantile regression is deferred to Chapter 4. Several approaches to inference are considered: Wald tests and related problems of direct estimation of the asymptotic covariance matrix, rank tests based on the dual quantile regression process, likelihood-ratio-type tests based on the value of the objective function under null and alternative models, and, finally, several resampling methods are introduced. The chapter concludes with a description of a small Monte Carlo experiment designed to evaluate and compare the foregoing methods.

3.1 THE FINITE-SAMPLE DISTRIBUTION OF REGRESSION QUANTILES

Suppose $Y_1, \ldots, Y_n$ are iid random variables with common distribution function F, and assume that F has a continuous density f in a neighborhood of

$\xi_\tau = F^{-1}(\tau)$ with $f(\xi_\tau) > 0$. The objective function of the τth sample quantile,

$$\hat{\xi}_\tau \equiv \inf_\xi \left\{ \xi \in \mathbb{R} | \sum_{i=1}^{n} \rho_\tau(Y_i - \xi) = \min! \right\},$$

is the sum of convex functions and, hence, is itself convex. Consequently, by the monotonicity of the gradient,

$$g_n(\xi) = \sum_{i=1}^{n} (I(Y_i < \xi) - \tau),$$

we have

$$\begin{aligned} P\{\hat{\xi}_\tau > \xi\} &= P\{g_n(\xi) < 0\} \\ &= P\left\{\sum I(Y_i < \xi) < n\tau\right\} \\ &= P\{B(n, F(\xi)) < n\tau\}, \end{aligned}$$

where $B(n, p)$ denotes a binomial random variable with parameters (n, p). Thus, letting $m = \lceil n\tau \rceil$ denote the smallest integer $\geq n\tau$, we may express the distribution function, $G(\xi) \equiv P\{\hat{\xi}_\tau \leq \xi\}$, of $\hat{\xi}(\tau)$, using the incomplete beta function, as

$$\begin{aligned} G(\xi) &= 1 - \sum_{k=m}^{n} \binom{n}{k} F(\xi)^k (1 - F(\xi))^{n-k} \\ &= n \binom{n-1}{m-1} \int_0^{F(\xi)} t^{m-1} (1-t)^{n-m} dt. \end{aligned}$$

Differentiating yields the density function for $\hat{\xi}(\tau)$;

$$g(\xi) = n \binom{n-1}{m-1} F(\xi)^{m-1} (1 - F(\xi))^{n-m} f(\xi). \tag{3.1}$$

This form of the density can be deduced directly by noting that the event $\{x < Y_{(m)} < x + \delta\}$ requires that $m - 1$ observations lie below x, $n - m$ lie above $x + \delta$, and one lies in the interval $(x, x + \delta)$. The number of ways that this arrangement can occur is

$$\frac{n!}{(m-1)!1!(n-m)!} = n \binom{n-1}{m-1},$$

and each arrangement has the probability $F(\xi)^{m-1}(1 - F(\xi))^{n-m}[F(\xi + \delta) - F(\xi))]$. Thus,

$$P\{x < Y_{(m)} < x + \delta\} = n \binom{n-1}{m-1} F(\xi)^{m-1} (1 - F(\xi))^{n-m} f(\xi)\delta + o(\delta^2),$$

and we obtain (3.1) by dividing both sides by δ and letting it tend to zero.

This approach may also be used to construct confidence intervals for ξ_τ of the form

$$P\{\hat{\xi}_{\tau_1} < \xi_\tau < \hat{\xi}_{\tau_2}\} = 1 - \alpha,$$

where τ_1 and τ_2 are chosen to satisfy

$$P\{n\tau_1 < B(n, \tau) < n\tau_2\} = 1 - \alpha.$$

In the case of continuous F, these intervals have the remarkable property that they are distribution free; that is, they hold irrespective of F. In the case of discrete F, closely related distribution-free *bounds* for $P\{\hat{\xi}_{\tau_1} < \xi_\tau < \hat{\xi}_{\tau_2}\}$ and $P\{\hat{\xi}_{\tau_1} \le \xi_\tau \le \hat{\xi}_{\tau_2}\}$ may be constructed.

The finite-sample density of the regression quantile estimator, $\hat{\beta}(\tau)$, may be derived under iid errors from subgradient condition (2.5) introduced in the previous chapter in a manner much like the derivation of the density in the one-sample case given earlier.

Theorem 3.1 (Koenker and Bassett 1978). *Consider the linear model*

$$Y_i = x_i^\top \beta + u_i \quad i = 1, \dots, n$$

with iid errors $\{u_i\}$ having a common distribution function F and strictly positive density f at $F^{-1}(\tau)$. Then the density of $\hat{\beta}(\tau)$ takes the form

$$g(b) = \sum_{h \in \mathcal{H}} P\{\xi_h(b) \in \mathcal{C}\} \big|X(h)\big| \prod_{i \in h} f\left(x_i^\top (b - \beta(\tau)\right) + F^{-1}(\tau)), \tag{3.2}$$

where $\xi_h(b) = \sum_{i \in \bar{h}} \psi_\tau(y_i - x_i b) x_i^\top X(h)^{-1}$ and $\mathcal{C}$ denotes the cube $[\tau - 1, \tau]^p$.

Proof. From Theorem 2.1, $\hat{\beta}(\tau) = b(h) \equiv (X(h))^{-1} Y(h)$ if and only if $\xi_h(b(h)) \in \mathcal{C}$. For any $b \in \mathbb{R}^p$, let $B(b, \delta) = b + [-\delta/2, \delta/2]^p$ denote the cube centered at b with edges of length δ and write

$$\begin{aligned} P\{\hat{\beta}(\tau) \in B(b, \delta)\} &= \sum_{h \in \mathcal{H}} P\{b(h) \in B(b, \delta), \xi_h(b(h)) \in \mathcal{C}\} \qquad (3.3) \\ &= \sum_{h \in \mathcal{H}} E I(b(h) \in B(b, \delta)) P\{\xi_h(b(h)) \in \mathcal{C} | Y(h)\}, \end{aligned}$$

where the expectation is taken with respect to the vector $Y(h)$. The preceding conditional probability is defined from the distribution of $\xi_h(b(h))$, which is a discrete random variable (taking on 2^{n-p} values for each $h \in \mathcal{H}$). As $\delta \to 0$, this conditional probability tends to $P\{\xi_h(b) \in \mathcal{C}\}$; this probability is independent of $Y(h)$.

Now, divide both sides by Volume$(B(b, \delta)) = \delta^p$, and let $\delta \to 0$ to put things in density form. The conditional probability tends to $P\{\xi_h(b) \in \mathcal{C}\}$, which no longer depends on $Y(h)$. The other factor tends to the joint density of the vector $X(h)^{-1}Y(h)$ which, since

$$f_{Y(h)}(y) = \prod_{i \in h} f\left(y_i - x_i^\top \beta\right),$$

can be written as

$$\begin{aligned} f_{(X(h))^{-1}Y(h)}(b) &= \left|X(h)\right| \prod_{i \in h} f\left(\left(X(h)b\right)_i - x_i^\top \beta(\tau) + F^{-1}(\tau)\right) \\ &= \left|X(h)\right| \prod_{i \in h} f\left(x_i^\top (b - \beta(\tau)) + F^{-1}(\tau)\right). \qquad (3.4) \end{aligned}$$

Note that h, for which $X(h)$ is singular, contributes nothing to density. The result now follows by reassembling the pieces. ■

Unfortunately, from a practical standpoint, the $\binom{n}{p}$ summands of (3.2) are not very tractable in most applications and, as in the least-squares theory, we must resort to asymptotic approximations for a distribution theory adaptable to practical statistical inference. In the next section, we try to provide a heuristic introduction to the asymptotic theory of quantile regression. A more detailed formal treatment of the asymptotic theory of quantile regression may be found in Chapter 4.

3.2 A HEURISTIC INTRODUCTION TO QUANTILE REGRESSION ASYMPTOTICS

The optimization view of the sample quantiles also affords an elementary approach to their asymptotic theory. Suppose $Y_1, \ldots, Y_n$ are iid from the distribution F and for some quantile $\xi_\tau = F^{-1}(\tau)$ assume that F has a continuous density f at ξ_τ with $f(\xi_\tau) > 0$. The objective function of the τth sample quantile,

$$\hat{\xi}_\tau \equiv \inf_{\xi}\{\xi \in \mathbb{R} | n^{-1} \sum \rho_\tau(Y_i - \xi) = \min!\},$$

as we have already noted, is the sum of convex functions and, hence, is itself convex. Consequently, its gradient

$$g_n(\xi) = n^{-1} \sum_{i=1}^{n} (I(Y_i < \xi) - \tau)$$

is monotone in ξ. Of course, when ξ equals one of the Y_i then this "gradient" needs the subgradient interpretation discussed earlier, but this is not crucial to the argument that follows.

The function g_n is clearly monotonically increasing in ξ and, by monotonicity, $\hat{\xi}_\tau$ is greater than ξ if and only if $g_n(\xi) < 0$, and so

$$\begin{aligned} P\{\sqrt{n}(\hat{\xi}_\tau - \xi_\tau) > \delta\} &= P\{g_n(\xi_\tau + \delta/\sqrt{n}) < 0\} \\ &= P\{n^{-1} \sum (I(Y_i < \xi_\tau + \delta/\sqrt{n}) - \tau) < 0\}. \end{aligned}$$

Thus, we can reduce the asymptotic behavior of $\hat{\xi}_\tau$ to a DeMoivre–Laplace central limit theorem problem in which we have a triangular array of Bernoulli random variables. The summands take the values $(1 - \tau)$ and $-\tau$ with probabilities $F(\xi_\tau + \delta/\sqrt{n})$ and $1 - F(\xi_\tau + \delta/\sqrt{n})$, respectively. Expanding F, we have

$$Eg_n(\xi_\tau + \delta/\sqrt{n}) = (F(\xi_\tau + \delta/\sqrt{n}) - \tau) \to f(\xi_\tau)\delta/\sqrt{n}$$

and

$$V(g_n(\xi_\tau + \delta/\sqrt{n})) = F(\xi_\tau + \delta/\sqrt{n})(1 - F(\xi_\tau + \delta/\sqrt{n}))/n \to \tau(1-\tau)/n,$$

and so we may set $\omega^2 = \tau(1-\tau)/f^2(\xi_\tau)$ and write

$$\begin{aligned} P\{\sqrt{n}(\hat{\xi}_\tau - \xi_\tau) > \delta\} \;&=\; P\left\{\frac{g_n(\xi_\tau + \delta/\sqrt{n}) - f(\xi_\tau)\delta/\sqrt{n}}{\sqrt{\tau(1-\tau)/n}} < -\omega^{-1}\delta\right\} \\ &\to 1 - \Phi(\omega^{-1}\delta). \end{aligned}$$

Therefore, we conclude that

$$\sqrt{n}(\hat{\xi}_\tau - \xi_\tau) \rightsquigarrow \mathcal{N}(0, \omega^2). \tag{3.5}$$

There are two influences on the precision of the τth sample quantile. The $\tau(1-\tau)$ effect tends to make $\hat{\xi}_\tau$ more precise in the tails, but this would be typically dominated by the effect of the density term, which tends to make $\hat{\xi}_\tau$ less precise in regions of low density.

Extending the foregoing argument to consider the limiting form of the joint distribution of several quantiles, set $\hat{\zeta}_n = (\hat{\xi}_{\tau_1}, \ldots, \hat{\xi}_{\tau_m})$ with $\zeta_n = (\xi_{\tau_1}, \ldots, \xi_{\tau_m})$ and we obtain (see Problem 3.1)

$$\sqrt{n}(\hat{\zeta}_n - \zeta) \rightsquigarrow \mathcal{N}(0, \Omega), \tag{3.6}$$

where Ω is an $m \times m$ matrix composed of the following elements:

$$(\omega_{ij}) = (\tau_i \wedge \tau_j - \tau_i\tau_j)/(f(F^{-1}(\tau_i))f(F^{-1}(\tau_j))).$$

This result is the starting point of the large-sample theory for finite linear combinations of order statistics.

3.2.1 Confidence Intervals for the Sample Quantiles

Given an estimate of the nuisance parameter, ω, one can easily make confidence interval estimates for the parameter $\xi(\tau)$ based on the asymptotic approximation given in (3.5). But there is a more direct route to confidence intervals. Suppose we were interested in testing the hypothesis $H_0 : \xi(\tau) = \xi_0(\tau)$. Under this null hypothesis, the statistic

$$Z_n = \sum_{i=1}^{n} I(Y_i - \xi_0(\tau))$$

is binomial, $B(n, \tau)$, and so a test could be based on Z_n, rejecting H_0, if

$$T_n = n^{-1} Z_n - \tau$$

was too large in absolute value. Of course, we need to be able to evaluate how large "too large" is. A nice feature of this test is that it requires no auxiliary estimation of nuisance parameters and the null distribution of the test statistic is independent of the original distribution, F, of the observations. Note that T_n is precisely our gradient $g_n(\xi_0(\tau))$ evaluated at the hypothesized parameter $\xi_0(\tau)$.

To convert our test based on T_n to a procedure for constructing a confidence interval, we need to look for the set

$$C_\alpha = \{\xi : T_n(\xi) \text{ does not reject at level } \alpha\}.$$

Because $T_n(\xi)$ is piecewise constant and monotone increasing, we are restricted to a finite number of distinct confidence levels. Such intervals take a particularly simple form – they are intervals between two order statistics:

$$C_\alpha = [Y_{(L)}, Y_{(U)}],$$

where $Y_{(i)}$ denotes the ith-order statistic from the sample: $Y_1, Y_2, \ldots, Y_n$. The choice of the integers L and U can be determined directly from the exact binomial theory of T_n or may be approximated in large samples as

$$\{L, U\} = n\tau \pm c_{\alpha/2}\sqrt{n\tau(1-\tau)},$$

where $c_{\alpha/2} = \Phi^{-1}(1 - \alpha/2)$. Zhou and Portnoy (1996) discuss extensions of these procedures for contructing prediction intervals for general quantile regression models. Prediction intervals based on estimated inter-quantile ranges are particularly well-suited to applications in which the predictive density is thought to be asymmetric. The rank test inversion procedures of Section 3.5.5 are also closely related to these procedures.

3.2.2 Quantile Regression Asymptotics with IID Errors

These results for the ordinary sample quantiles in the one-sample model generalize nicely to the classical linear regression model,

$$y_i = x_i^\top \beta + u_i,$$

with iid errors $\{u_i\}$. Suppose that $\{u_i\}$ have a common distribution function F with associated density f, with $f(F^{-1}(\tau_i)) > 0$ for $i = 1, \ldots, m$, and $n^{-1}\sum x_i x_i^\top \equiv Q_n$ converges to a positive definite matrix Q_0. Then the joint asymptotic distribution of the mp-variate quantile regression estimators $\hat{\zeta}_n = (\hat{\beta}_n(\tau_1)^\top, \ldots, \hat{\beta}_n(\tau_m)^\top)^\top$ takes the form

$$\sqrt{n}(\hat{\zeta}_n - \zeta) = (\sqrt{n}(\hat{\beta}_n(\tau_j) - \beta(\tau_j)))_{j=1}^m \rightsquigarrow \mathcal{N}\big(0, \Omega \otimes Q_0^{-1}\big). \qquad (3.7)$$

A formal discussion of this result is deferred until Section 4.3. In the iid error regression setting, the form of the $\beta(\tau_j)$ is particularly simple. The conditional quantile planes are parallel, and so, presuming that the first coordinate of β corresponds to the "intercept" parameter, we have $\beta(\tau) = \beta + \xi_\tau e_1$, where $\xi_\tau = F^{-1}(\tau)$ and e_1 is the first unit basis vector of $\mathbb{R}^p$. Because Ω takes the same form as in the one-sample setting, many of the classical results on L-statistics can be directly carried forward to iid error regression using this result. This result, which is essentially the content of Theorem 4.1 of Koenker and Bassett (1978), affords considerable scope for Wald-type inference in the quantile regression setting. Hypotheses that may be formulated as linear restrictions of the vector ζ are immediately subject to test using the limiting normal theory and its χ^2 adaptations.

3.2.3 Quantile Regression Asymptotics in Non-IID Settings

The classical iid error linear regression model yields a particularly simple form for the limiting distribution of the quantile regression estimator $\hat{\beta}(\tau)$. However, it might be argued that, in the location shift form of the iid error model, quantile regression is really superfluous; in this case any reasonable estimator of the conditional central tendency of the response, given the covariates, is perfectly adequate. In non-iid error settings like the conditional location-scale model introduced in Section 2.5, the asymptotic theory of $\hat{\beta}(\tau)$ is somewhat more complicated. As we shall show in Section 4.3, the limiting covariance matrix of $\sqrt{n}(\hat{\beta}(\tau) - \beta(\tau))$ takes the form of a Huber (1967) sandwich:

$$\sqrt{n}(\hat{\beta}(\tau) - \beta(\tau)) \rightsquigarrow \mathcal{N}\left(0, \tau(1-\tau)H_n^{-1} J_n H_n^{-1}\right),$$

where

$$J_n(\tau) = n^{-1} \sum_{i=1}^{n} x_i x_i^\top$$

and

$$H_n(\tau) = \lim_{n\to\infty} n^{-1} \sum_{i=1}^{n} x_i x_i^\top f_i(\xi_i(\tau)).$$

The term $f_i(\xi_i(\tau))$ denotes the conditional density of the response, y_i, evaluated at the τth conditional quantile. In the iid case, these f_is are identical, and the sandwich collapses to the expression we have already considered. We are then faced with the relatively simple problem of estimating a density, or its reciprocal, at a point. However, in the non-iid case, we face a somewhat more challenging task.

The preceding approach may be extended easily to the problem of estimating asymptotic covariance matrices for distinct vectors of quantile regression parameters. In these cases we would like to estimate, the asymptotic covariance matrix for $\hat{\beta}(\tau_1), \ldots, \hat{\beta}(\tau_m)$ has blocks

$$\begin{aligned} &\mathrm{Acov}(\sqrt{n}(\hat{\beta}(\tau_i) - \beta(\tau_i)), \sqrt{n}(\hat{\beta}(\tau_j) - \beta(\tau_j))) \\ &\quad = [\tau_i \wedge \tau_j - \tau_i\tau_j] H_n(\tau_i)^{-1} J_n H_n(\tau_j)^{-1}, \end{aligned}$$

where i and j run from 1 to m. Thus, Wald tests, like the heteroscedasticity tests described earlier, that involve linear restrictions across several quantile regression parameter vectors can be easily carried out with the same machinery. It should be emphasized that this approach is computationally extremely efficient and, thus, is particularly attractive for large problems where bootstrapping and the rank test inversion approaches (discussed later) are computationally demanding.

3.3 WALD TESTS

The classical theory of linear regression *assumes* that the conditional quantile functions of the response variable, y, given covariates x, are all parallel to one another, implying that the slope coefficients of distinct quantile regressions will be identical. Covariate effects shift the location of the response distribution but do not change its scale or shape. In applications, however, as we have seen, quantile regression slope estimates often vary considerably across quantiles, and so an immediate and fundamental problem of inference in quantile regression involves testing for equality of slope parameters across quantiles. Some simple tests designed for this purpose were suggested by Koenker and Bassett (1982a). We will first introduce a simple version of these tests for the two-sample problem and then generalize to more complicated settings.

3.3.1 Two-Sample Tests of Location Shift

For the two-sample problem, these Wald tests correspond to tests of equality between the interquantile ranges of the two samples. Thus, they may be considered to be tests of homogeneity of scale or tests for heteroscedasticity. Consider the two-sample model

$$Y_i = \alpha_1 + \alpha_2 x_i + u_i,$$

where $x_i = 0$ for n_1 observations in the first sample and $x_i = 1$ for n_2 observations in the second sample. The τth regression quantile estimate of the "slope" parameter α_2 in this model is simply the difference between the τth sample quantiles of the two samples (see Problem 2.2). Thus, a test of the equality of the slope parameters across quantiles τ_1 and τ_2 is just a test of the hypothesis

$$\begin{aligned}\alpha_2(\tau_2) - \alpha_2(\tau_1) &= (Q_2(\tau_2) - Q_1(\tau_2)) - (Q_2(\tau_1) - Q_1(\tau_1))\\ &= (Q_2(\tau_2) - Q_2(\tau_1)) - (Q_1(\tau_2) - Q_1(\tau_1))\\ &= 0,\end{aligned}$$

that is, that the $(\tau_2 - \tau_1)$ interquantile ranges are identical for the two samples. By (3.7) the asymptotic variance of $\hat{\alpha}_2(\tau_2) - \hat{\alpha}_2(\tau_1)$ is given by

$$\sigma^2(\tau_1, \tau_2) = \left[\frac{\tau_1(1-\tau_1)}{f^2(\xi_1)} - 2\frac{\tau_1(1-\tau_2)}{f(\xi_1)f(\xi_2)} + \frac{\tau_2(1-\tau_2)}{f^2(\xi_2)}\right]\left[\frac{n}{nn_2 - n_2^2}\right],$$

where $\xi_i = F^{-1}(\tau_i)$ and a test of the null hypothesis can be based on the asymptotic normality of the statistic,

$$T_n = (\hat{\alpha}_2(\tau_2) - \hat{\alpha}_2(\tau_1))/\hat{\sigma}(\tau_1, \tau_2).$$

Of course, it is obvious from the form of $\sigma^2(\tau_1, \tau_2)$ that it is necessary to estimate the nuisance parameters, $1/f(F^{-1}(\tau_1))$ and $1/f(F^{-1}(\tau_2))$, a topic which will be deferred to Section 3.4.1.

A variety of other tests are immediately suggested by this construction. For example, we can test that the symmetry of the response distribution under the treatment with the following hypothesis:

$$H_0: \qquad \alpha_2\left(\frac{1}{2}+\delta\right) - \alpha_2\left(\frac{1}{2}-\delta\right) - 2\alpha_2\left(\frac{1}{2}\right) = 0$$

for some $\delta \in (0, \frac{1}{2})$. The hypothesis

$$H_0: \qquad \alpha_1\left(\frac{1}{2}+\delta\right) - \alpha_1\left(\frac{1}{2}-\delta\right) - 2\alpha_1\left(\frac{1}{2}\right) = 0$$

focuses attention solely on the symmetry of the control distribution. Combining the two linear restrictions yields a test of the joint hypothesis that both the control and treatment distributions are symmetric. This leads us to the general linear hypotheses of the next section.

3.3.2 General Linear Hypotheses

More general hypotheses are easily accommodated by the Wald approach. For example, as Koenker and Bassett (1982b), we may consider a general linear hypothesis on the vector $\zeta = (\beta(\tau_1)^\top, \ldots, \beta(\tau_m)^\top)^\top$ of the form

$$H_0: \ R\zeta = r$$

and test statistic

$$T_n = n(R\hat{\zeta} - r)^\top [RV^{-1}R^\top]^{-1}(R\hat{\zeta} - r),$$

where V_n is the $mp \times mp$ matrix with ijth block

$$V_n(\tau_i, \tau_j) = [\tau_i \wedge \tau_j - \tau_i \tau_j] H_n(\tau_i)^{-1} J_n(\tau_i, \tau_j) H_n(\tau_j)^{-1}, \qquad (3.8)$$

as at the end of the previous section. The statistic T_n is asymptotically χ_q^2 under H_0, where q is the rank of the matrix R. This formulation accommodates a wide variety of testing situations, from simple tests on a single quantile regression coefficient to joint tests involving several covariates and several distinct quantiles. Thus, for example, we might test for the equality of several slope coefficients across several quantiles; such tests provide a robust alternative to conventional least-squares-based tests of heteroscedasticity because they can be constructed to be insensitive to outlying response observations. The same formulation can, of course, be adopted to accommodate nonlinear hypotheses on the vector ζ by interpreting H_0 as the Jacobian of the nonlinear hypothesis. Newey and Powell (1987) discuss tests for symmetry employing this approach.

3.4 ESTIMATION OF ASYMPTOTIC COVARIANCE MATRICES

In this section we consider the problem of estimating the asymptotic covariance matrix. We begin with the iid error setting and then consider more general conditions.

3.4.1 Scalar Sparsity Estimation

It is a somewhat unhappy fact of life that the asymptotic precision of quantile estimates in general, and quantile regression estimates in particular, depend on the reciprocal of a density function evaluated at the quantile of interest – a quantity Tukey (1965) has termed the "sparsity function" and Parzen (1979) calls the quantile-density function. It is perfectly natural that the precision of quantile estimates should depend on this quantity because it reflects the density of observations near the quantile of interest; if the data are very sparse at the quantile of interest they will be difficult to estimate. On the other hand, when the sparsity is low, so that there are many observations near the quantile of interest, then the quantile will be more precisely estimated. Thus, to estimate the precision of the τth quantile regression estimate directly, the nuisance quantity

$$s(\tau) = [f(F^{-1}(\tau))]^{-1}$$

must be estimated, and this leads us into the realm of density estimation and smoothing. We shall see that it is possible to pull ourselves out of this swamp by the bootstraps, or related statistical necromancy, but we defer the exploration of these strategies a bit and begin by exploring the more direct approach of estimating the asymptotic covariance matrix.

Differentiating the identity $F(F^{-1}(t)) = t$, we find that the sparsity function is simply the derivative of the quantile function; that is,

$$\frac{d}{dt}F^{-1}(t) = s(t).$$

Therefore, just as differentiating the distribution function F yields the density function f, differentiating the quantile function F^{-1} yields the sparsity function s. It is therefore natural, following Siddiqui (1960), to estimate $s(t)$ by using a simple difference quotient of the empirical quantile function:

$$\hat{s}_n(t) = [\hat{F}_n^{-1}(t + h_n) - \hat{F}_n^{-1}(t - h_n)]/2h_n,$$

where $\hat{F}^{-1}$ is an estimate of F^{-1} and h_n is a bandwidth that tends to zero as $n \to \infty$. Choice of the bandwidth is discussed in Section 4.10.1.

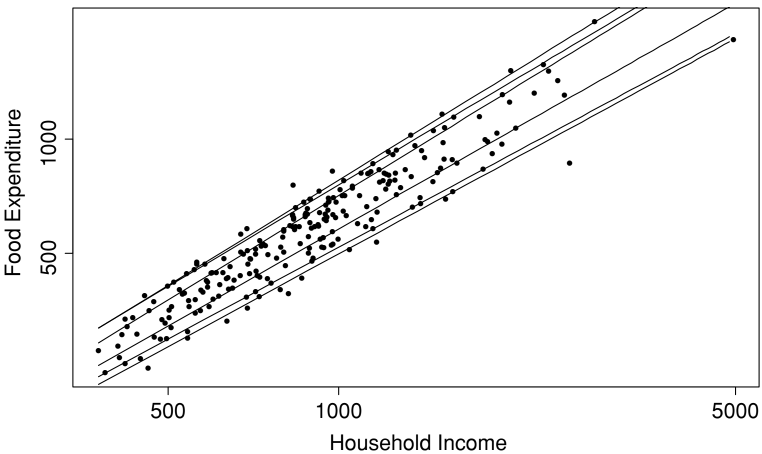

Figure 3.1. Engel curves for food. This figure plots data taken from Ernst Engel's (1857) study of households' expenditure on food versus annual income. The data consist of 235 observations on European working-class households. Superimposed on the plot are six estimated quantile regression lines corresponding to the quantiles $\tau \in \{0.05, 0.1, 0.25, 0.75, 0.9, 0.95\}$.

To illustrate the application of this approach, consider the data depicted in Figure 3.1. This is a classical data set in economics and is based on 235 budget surveys of 19th century working-class households. Household expenditure on food is measured vertically, and household income is measured horizontally. The data were originally presented by Ernst Engel (1857) to support his hypothesis that food expenditure constitutes a declining share of household income. Following established custom, we have transformed both variables to the log-scale, so that a slope less than unity is evidence for Engel's law. We have superimposed six estimated linear conditional quantile functions for these data. The fitted lines look roughly parallel, but we might like a formal test of the hypothesis of equality of slopes. Focusing on the inner two lines, which represent the fit for the first ($\tau = 0.25$) and third ($\tau = 0.75$) quartiles, we find that the difference in the slopes is

$$\hat{\beta}_2(3/4) - \hat{\beta}_2(1/4) = 0.915 - 0.849 = 0.0661.$$

In Figure 3.2 we illustrate the function $\hat{Q}_Y(\tau|\bar{x})$ for this data set. The vertical scale is the natural logarithm of food expenditure. The dotted lines forming triangles illustrate the estimation of the sparsity function at the first and third quartiles. The Hall–Sheather bandwidth for both estimates is 0.097, yielding sparsity estimates of $\hat{s}(1/4) = 0.543$ and $\hat{s}(3/4) = 0.330$. The lower diagonal element of $(X^\top X)^{-1}$ is 0.022, and so the test statistic for the equality of the two slopes is 1.93, which has a p-value of 0.03 for a one-tailed test of the hypothesis of equality of the slopes. This result offers very weak evidence of increasing dispersion in the logarithm of food expenditure with income, a finding that may

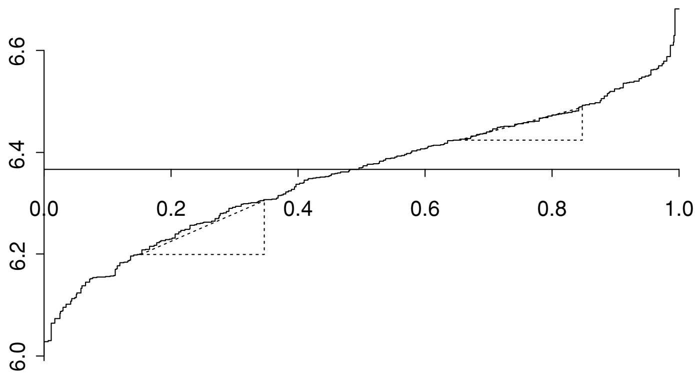

Figure 3.2. Sparsity estimation for the Engel data. This figure plots $\hat{Q}_Y(\tau|\bar{x}) = \bar{x}'\hat{\beta}(\tau)$ for the Engel data. The vertical scale is logarithmic in expenditure on food. The estimation of the sparsity function by the Siddiqui method is illustrated by the dotted triangles that depict the difference quotient estimator of the sparsity at the first and third quartiles. The estimate of the sparsity is given by the slope of the hypotenuse of the triangles. The Hall–Sheather bandwidth is 0.097 for this example.

seem surprising in view of Figure 3.1. In large samples, the formal statistical significance of such tests is extremely common. The substantive significance of such heteroscedasticity is, of course, completely open to question, too. It would be very surprising if such a small difference in slopes could be viewed as "economically significant."

There are obviously many other possible approaches to the estimation of the sparsity parameter. Welsh (1987) considers a kernel approach that may be interpreted as a weighted average of Siddiqui estimates in which those with narrow bandwidth are given greater weight. Another approach is suggested by simply fitting a local polynomial to $\hat{F}^{-1}(t)$ in the neighborhood of τ and using the slope of this fitted function at τ as an estimate of sparsity. Koenker and Bassett (1982a) employed the histospline methods of Boneva, Kendall, and Stefanov (1971) to estimate the sparsity function.

3.4.2 Covariance Matrix Estimation in Non-IID Settings

Two approaches to the estimation of the matrix H_n will be described. One is the natural extension of sparsity estimation methods described earlier, suggested by Hendricks and Koenker (1991). The other, which is based on kernel density estimation ideas, was proposed by Powell (1991).

The Hendricks–Koenker Sandwich

Provided that the τth conditional quantile function of $y|x$ is linear, then for $h_n \to 0$ we can consistently estimate the parameters of the $\tau \pm h_n$ conditional quantile functions by $\hat{\beta}(\tau \pm h_n)$. And the density $f_i(\xi_i)$ can thus be estimated by the difference quotient

$$\hat{F}_i(\xi_i(\tau)) = 2h_n / x_i^\top(\hat{\beta}(\tau + h_n) - \hat{\beta}(\tau - h_n)),$$

using the same choice of bandwidth mentioned earlier. Substituting this estimate in the expression for H_n preceding yields an easily implementable estimator for the asymptotic covariance matrix of $\hat{\beta}(\tau)$ in the non-iid error model. Note that the matrix J_n involves no nuisance parameters and thus is immediately computable.

A potential difficulty with the proposed estimator $\hat{F}_i(\xi_i(\tau))$ is that there is no guarantee of positivity for every observation in the sample. Indeed, as we have already seen, the quantity

$$d_i = x_i^\top(\hat{\beta}(\tau + h_n) - \hat{\beta}(\tau - h_n))$$

is *necessarily* positive only at $x = \bar{x} = n^{-1} \sum x_i$. Nevertheless, in practice we find that problems due to "crossing" of the estimated conditional quantile planes occur only infrequently and in the most extreme regions of the design space. In the implementation of this approach, we simply replace $\hat{F}_i$ by its positive part; that is, we use

$$\hat{F}_i^+ = \max\{0, 2h_n/(d_i - \varepsilon)\},$$

where $\varepsilon > 0$ is a small tolerance parameter intended to avoid dividing by zero in the (rare) cases in which $d_i = 0$ because the ith observation is basic at both $\tau \pm h_n$.

The Powell Sandwich

Powell has suggested an alternative, and in some respects even simpler, way to estimate the quantile regression sandwich. Noting that in estimating the matrix $H_n(\tau)$ we are really after a matrix weighted density estimator, he proposes a kernel estimator of the form

$$\hat{H}_n(\tau) = (nh_n)^{-1} \sum K(\hat{u}_i(\tau)/h_n) x_i x_i^\top,$$

where $\hat{u}_i(\tau) = y_i - x_i^\top \hat{\beta}(\tau)$ and h_n is a bandwidth parameter satisfying $h_n \to 0$ and $\sqrt{n} h_n \to \infty$. He shows that, under certain uniform continuity requirements on the f_i, $\hat{H}_n(\tau) \to H_n(\tau)$ in probability. In practice, of course, there remains a number of nettlesome questions about the choice of the kernel function K and the bandwidth parameter h_n. We will have more to say about this later when we describe a small Monte Carlo experiment designed to compare the performance of various confidence interval strategies for quantile regression.

Powell (1991) uses

$$\hat{H} = \frac{1}{2nc_n} \sum I(|\hat{u}_i| < c_n) x_i x_i^\top,$$

where $c_n \to 0$ and $\sqrt{n}c_n \to \infty$. In the R implementation of this approach, I use the default bandwidth

$$c_n = \kappa(\Phi^{-1}(\tau + h_n) - \Phi^{-1}(\tau - h_n)),$$

where κ is a robust estimate of scale. The bandwidth $h_n = \mathcal{O}(n^{-1/3})$, so $c_n = \mathcal{O}(n^{-1/3})$ and Powell's conditions are satisfied. Note that, as $h_n \to 0$,

$$\frac{\Phi^{-1}(\tau + h_n) - \Phi^{-1}(\tau - h_n)}{2h_n} \to \frac{1}{\phi(\Phi^{-1}(\tau))}.$$

3.5 RANK-BASED INFERENCE

The Wald approach to inference starts from the vantage point of the alternative hypothesis and asks the following question: Is it plausible that certain linear restrictions hold? Alternatively, we may consider starting at the null hypothesis and asking: Is there a direction to move in the space of alternative hypotheses that leads to more plausible estimates? This is the "score" approach of Rao (1947). Application of the score approach to quantile regression inference results in a remarkable confluence of statistical theory: score tests for quantile regression are, in effect, a class of generalized rank tests for the linear regression model. This observation was first made by Gutenbrunner and Jurečková (1992), who linked the dual quantile regression process to the classical Hájek (1965) rankscore functions, and was elaborated by Gutenbrunner, Jurečková, Koenker, and Portnoy (1993).

The works of Spearman (1904), Hotelling and Pabst (1936), Friedman (1937), and Kendall (1938) are generally credited with initiating the rank-based approach to statistical inference. The appearance of the work of Wilcoxon (1945), Mann and Whitney (1947), and Hoeffding (1948) on tests for location shift based on ranks, and the subsequent analysis of these tests by Hodges and Lehmann (1956) and Chernoff and Savage (1958), firmly established the subject as a precocious challenger to classical likelihood-based methods of inference. The rigorous reformulation of the asymptotic theory of rank tests introduced by Hájek (1965) and developed by Hájek and Šidák (1967) and others, in conjunction with the emergence of robustness as a major theme of statistical research in the late 1960s gave a significant impetus to the growth of rank-based methods.

3.5.1 Rank Tests for Two-Sample Location Shift

As an introduction to rank tests in an elementary setting, we may begin by comparing the performance of the Wilcoxon–Mann–Whitney test for location shift with the classical two-sample t-test. Let $X_1, \ldots, X_n$ be a random sample of "control" observations from the distribution function $F(x)$ and let $Y_1, \ldots, Y_m$ denote a random sample of "treatment" observations from $F(x - \theta)$. We assume at the outset that the treatment constitutes a pure location shift. Suppose we wish to test the hypothesis of "no treatment effect," $H_0 : \theta = 0$, against the alternative, $H_1 : \theta > 0$.

If F were known to be Gaussian, we would immediately compute the sample means $\bar{X}_n$ and $\bar{Y}_m$ and then compute the test statistic

$$T = (\bar{X}_n - \bar{Y}_m)/s,$$

where $s^2 = \sigma^2(n^{-1} + m^{-1})$, replacing σ^2 by

$$\hat{\sigma}^2 = (n + m - 2)^{-1}\left[\sum_{i=1}^{n}(X_i - \bar{X}_n)^2 + \sum_{i=1}^{m}(Y_i - \bar{Y}_m)^2\right]$$

if necessary. If T exceeded $\Phi^{-1}(1 - \alpha)$, we would reject H_0 at level α if σ^2 were known, replacing the normal critical value by its corresponding t_{n+m-2} value if σ^2 needed to be estimated.

Mann and Whitney (1947) proposed an alternative to the two-sample t-test is based on the statistic

$$S_1 = \sum_{i=1}^{m}\sum_{j=1}^{n} I(Y_i > X_j).$$

We simply count the number of pairs of observations – one from each sample – for which the treatment observation exceeds the control observation. If S_1 is large, it suggests that treatment observations are generally larger than controls and H_0 should be rejected.

How is this connected to ranks? Wilcoxon (1945) suggests an alternative formulation: pool the two samples and compute the rank of each observation in the pooled sample. Let S_0 denote the sum of the treatment observation ranks; again, if S_0 is large, it suggests rejection of H_0. It is easy to show that, in the absence of ties,

$$S_1 = S_0 - m(m + 1)/2,$$

and so the Mann–Whitney and Wilcoxon forms of the test are actually equivalent. Efron (1967) suggests a very nice interpretation of a normalized version of the Wilcoxon statistic as an estimate of the probability associated with the region over which $G(x)$ exceeds $F(x)$. One of the most attractive features of this test is that its null distribution is independent of the form of F generating the original observations. To see this, let $Z_1, \ldots, Z_n$ denote $X_1, \ldots, X_n$ and let $Z_{n+1}, \ldots, Z_{n+m}$ denote $Y_1, \ldots, Y_m$. Under H_0, the Z_i are iid. Thus, for any permutation $(i_1, \ldots, i_{n+m})$ of $(1, \ldots, n + m)$, $(Z_{i_1}, \ldots, Z_{i_{\mathrm{n+m}}})$ has the same distribution as $(Z_1, \ldots, Z_{n+m})$. Therefore, in the absence of "ties,"

$$\{R_1 = r_1, \ldots, R_{n+m} = r_{n+m}\} = \{Z_{i_1} < \cdots < Z_{i_{n+m}}\},$$

where R_i is the rank of the ith observation in the pooled sample and $r_{i_j} \equiv j$. It follows that the $(n + m)!$ possible events $\{R_1 = r_1, \ldots, R_{n+m} = r_{n+m}\}$ are equally likely. Because S_0 is a function solely of these ranks, its distribution under H_0 is also independent of F. For modest n, m we can compute exact critical values for the test based on combinatorial considerations (see, e.g., Lehmann, 1974, for details). For large n, m we can rely on the fact that, under H_0,

$$\tilde{S}_1 = (S_1 - \mu_1)/\sigma_1 \rightsquigarrow \mathcal{N}(0, 1),$$

where $\mu_1 = mn/2$ and $\sigma_1^2 = nm(n + m - 2)/12$. Even for modest sample sizes this approximation is quite good.

It is tempting to think that the main reason for preferring the Wilcoxon test to the t-test is the fact that it has a guaranteed probability of Type I error for quite arbitrary distributions of F, whereas the exact theory of the t-statistic depends on the normality of the observations. However, as long as F has a finite second moment and n and m are reasonably large, the critical values of the t distribution are also reasonably accurate for the t-statistic. It is power considerations that constitute the most compelling case for the Wilcoxon test under non-Gaussian conditions.

Exact power comparisons are rather impractical, but asymptotic results are very revealing. In the late 1940s Pitman proposed considering sequences of local alternatives of the form

$$H_n : \theta_n = \theta_0/\sqrt{n}$$

for precisely this problem, and showed that, at the normal model $F = \Phi$, the limiting ratio of sample sizes required to achieve the same size and power with the Wilcoxon and the t-test is $3/\pi \approx 0.955$. This ratio takes the general form

$$\text{ARE} = \frac{12\sigma^2(F)}{\omega^2(F)},$$

where $\sigma^2(F)$ denotes the variance of the distribution F and

$$\omega^2(F) = \int_0^1 f(F^{-1}(t))dt = \int_{-\infty}^{\infty} f^2(x)dx,$$

which is usually referred to as the asymptotic relative efficiency, or ARE, of the two tests. It provides a natural measure of their relative performance. If, for example, at the normal model, 1000 observations are required to achieve power 0.95 at level 0.05 for a given alternative with the Wilcoxon test, this implies that the same power and level would be achievable with the t-test with roughly 955 observations. Therefore, transforming to ranks has "wasted" about 5% of the observations. This loss of information is hardly surprising because the optimality of the t-test at the normal model is a cornerstone of statistical thinking.

What happens in non-Gaussian situations? Table 3.1 presents the asymptotic relative efficiency of the Wilcoxon test relative to the t-test in the location shift problem for eight well-known Fs.

The striking feature of this table is not the modest loss of information at the normal model but the enormous gains achieved by transforming to ranks in

Table 3.1. *ARE of Wilcoxon and t-tests of location shift*

F	Normal	Uniform	Logistic	Exp	t_3	Exp	LNormal	t_2
ARE	.955	1.0	1.097	1.5	1.9	3	7.35	∞

certain non-Gaussian situations. We would need 50% more observations if we foolishly used the t-test instead of the Wilcoxon test at the double exponential distribution, and *seven times* as many observations at the lognormal distribution. In the latter case, note that, if we were clever enough to do the log transformation to get back to normality, the Wilcoxon test is unaffected, because ranks are invariant to monotone transformations. But the transformation to ranks achieves most of what we could have achieved had we known that the log transformation was appropriate. For extremely heavy-tailed distributions, the advantage for the rank test is even more pronounced.

The message of Table 3.1 is a familiar one from robust statistics: if we are willing to pay a small insurance premium (5% efficiency loss) at the Gaussian model we can protect ourselves against the extremely poor performance of least-squares-based methods in heavy-tailed situations. The 5% premium seems very modest, particularly in view of the huge improvement in the log-normal case, but one may wonder whether there are other plausible distributions for which the asymptotic relative efficiency of the two tests is even worse than 0.955 in the Gaussian case. This question was answered originally by Hodges and Lehmann (1956). To find the f that minimizes ARE, one must minimize the L_2 norm of the density subject to the constraint that the density has variance one. This turns out to be isomorphic to a standard problem in kernel density estimation. The least favorable density turns out to be the one generating the Epanechnikov kernel, and this solution gives a lower bound to the ARE of about 0.864. Thus, even in non-Gaussian cases, the t-test can never be more than about 15% better than the Wilcoxon test of location shift, but *can be arbitrarily worse* as indicated in the table.

3.5.2 Linear Rank Statistics

A more general approach to two-sample rank tests is suggested by considering the distribution of the ranks when the Xs are generated from $f(x)$ and the Ys from $f(y-\theta)$. Let R denote the vector of ranks from the combined sample $Z = (X_1, \ldots, X_n, Y_1, \ldots, Y_m)$:

$$\begin{aligned} f_R(r;\theta) &= \int_{z_j = z_{(r_j)}} \prod_{k=1}^{n} f(z_k) \prod_{k=n+1}^{n+m} f(z_k - \theta) dz \\ &= \int_{\xi_1 < \ldots < \xi_{n+m}} \prod_{j=1}^{n+m} f(\xi_{r_j}) \prod_{k=1}^{m} \frac{f(\xi_{r_k} - \theta)}{f(\xi_{r_k})} d\xi, \end{aligned}$$

where $\xi_{r_j} = z_{(r_j)}$, $j = 1, \ldots, n+m$. But the density of the order statistics $Z_{(1)}, \ldots Z_{(n+m)}$ when $Z_1, \ldots Z_{n+m}$ are iid with density $f(z)$ is $(n+m)! \prod f(\xi_{r_k})$, and so

$$f_R(r;\theta) = ((n+m)!)^{-1} E_0 \prod_{k=1}^{m} \frac{f(\xi_{r_k} - \theta)}{f(\xi_{r_j})},$$

where E_0 designates that the expectation is taken with respect to the density $f(x)$, that is, under the null hypothesis that $\theta = 0$. For any fixed alternative θ, the optimal rank test could be computed as the likelihood ratio based on this expression for the density. A more practical option involves computing a locally most powerful test based on the score function

$$\frac{\partial \log f_R(R;\theta)}{\partial \theta} = -\sum_{j=1}^{n} E_0 \frac{f'\big(Z_{(R_j)}\big)}{f(Z_{(R_j)})}.$$

To illustrate, suppose f is logistic, so that

$$f(z) = e^z/(1+e^z)^2 = F(z)(1 - F(z)).$$

Then, $-f'(x)/f(x) = 2F(x) - 1$ and $F(Z(i)) = i/(n+m+1)$, and so the locally most powerful rank test of $H_0 : \theta = 0$ is based on the sum of the ranks and hence *is* the Wilcoxon–Mann–Whitney test. In contrast, if f were Gaussian, so that $-f'(x)/f(x) = x$, we may approximate the optimal rank test by using the statistic

$$T_n = \sum_{i=1}^{n} \Phi^{-1}(R_i/(n+m+1))$$

proposed by van der Waerden. Alternative approximations for $EZ_{(i)}$ with Z_i Gaussian have also been suggested (see, e.g., Hettmansperger, 1984). In general the approximate test statistic

$$T_n = \sum_{i=1}^{m} f'(F^{-1}(R_i/(n+m+1)))/f(F^{-1}(R_i/(n+m+1)))$$

may be used since $Z_{(i)} = F^{-1}(U_{(i)})$ where $U_1, \ldots, U_{n+m}$ are iid uniform on (0, 1), and $EU_{(i)} = i/(n+m+1)$.

3.5.3 Asymptotics of Linear Rank Statistics

The monograph of Hájek and Šidák (1967) constituted a complete reappraisal of the theory of rank statistics and provided an elegant general approach to the study of the asymptotic theory of linear rank statistics. For any sample $\{Y_1, Y_2, \ldots, Y_n\}$, and associated ranks $\{R_1, R_2, \ldots, R_n\}$, Hájek and Šidák introduced the rank-generating functions:

$$\hat{a}_i(t) = \begin{cases} 1 & \text{if } t \le (R_i - 1)/n \\ R_i - tn & \text{if } (R_i - 1)/n \le t \le R_i/n \\ 0 & \text{if } R_i/n \le t. \end{cases} \tag{3.9}$$

These functions "generate the ranks" of the Ys in the sense that, integrating with respect to Lebesgue measure,

$$\hat{b}_i = \int_0^1 \hat{a}_i(t)\, dt = (R_i - 1/2)/n,$$

and so the $\hat{b}_i$s *are* the ranks normalized to lie in the (0, 1) interval. More general notions of "ranks" may be obtained by replacing Lebesgue measure by alternative score functions $\varphi(t)$. For example, $\varphi(t) = 1/2sgn(t - 1/2)$ generates sign scores:

$$\hat{b}_i = \int \hat{a}_i(t)\, d\varphi(t) = \hat{a}_i(1/2) - 1/2 = \begin{cases} +1/2 & \text{if } R_i \geq n/2 + 1 \\ 0 & \text{otherwise} \\ -1/2 & \text{if } R_i \leq n/2. \end{cases}$$

The invariance of the ranks to monotone transformations of the Y_is means that the R_is may also be viewed as the ranks of the uniform random sample $\{U_1, \ldots, U_n\}$ with $U_i = F(Y_i)$, and the rank-generating functions $\hat{a}_i(t)$ may be seen as replacing the indicator functions $I(Y_i > F^{-1}(t)) = I(U_i > t)$ by the smoother "trapezoidal" form given by (3.9). The rank-generating functions behave like an empirical process as the following result shows.

Theorem 3.2 (Hájek and Šidák 1967, Theorem V.3.5). *Let* $c_n = (c_{11}, c_{21}, \ldots, c_{nn})$ *be a triangular array of real numbers satisfying*

$$\max(c_i - \bar{c})^2 / \sum_{i=1}^{n} (c_i - \bar{c})^2 \to 0 \tag{3.10}$$

and assume that $\{Y_1, \ldots., Y_n\}$ *constitute a random sample from an absolutely continuous distribution* F. *Then the process*

$$Z_n(t) = \left[\sum_{i=1}^{n} (c_i - \bar{c})^2\right]^{-1/2} \sum_{i=1}^{n} (c_i - \bar{c}) \hat{a}_i(t)$$

converges weakly to a Brownian bridge process on $C[0, 1]$.

In the two-sample problem, the c_{in}s may be taken simply as the indicator of which sample the observations belong to, and the "Lindeberg condition," (3.10), is satisfied as long as n_1/n stays bounded away from 0 and 1. A limiting normal theory for a broad class of linear rank statistics of the form

$$S_n = \left[\sum (c_{ni} - \bar{c}_n)^2\right]^{-1/2} \sum (c_{ni} - \bar{c}_n) \hat{b}_i,$$

where $\hat{b}_i = \int \hat{a}_i(t) d\varphi(t)$, is immediate. In particular, for any square integrable $\varphi : [0, 1] \to \mathbb{R}$, we have the linear representation

$$S_n = \left[\sum (c_{ni} - \bar{c}_n)^2\right]^{-1/2} \sum (c_{ni} - \bar{c}_n) \varphi(U_i) + o_p(1). \tag{3.11}$$

The score function φ can be chosen to target a variety of hypotheses, notably that the two samples differ by a location or scale shift. It follows that S_n is asymptotically Gaussian under the null with mean 0 and variance $A^2(\varphi) = \int (\varphi(t) - \bar{\varphi})^2 dt$, where $\bar{\varphi} = \int \varphi(t)dt$. Behavior under sequences of local alternatives can be studied using standard contiguity results.

Thus, for example, in the two-sample location shift model with local alternatives $H_n : \theta_n = \theta/\sqrt{n}$, we have S_n asymptotically Gaussian with mean $\omega(\varphi, F)(\sum(c_{in} - \bar{c}_n)^2)^{1/2}\theta$ and variance $A^2(\varphi)$, where

$$\omega^2(\varphi, F) = \int f(F^{-1}(t))d\varphi(t).$$

In the Wilcoxon case, φ is Lebesgue measure, and so $A^2(\varphi) = 1/12$ and $\omega^2 = \int f^2(x)dx$. Evaluating this expression yields Table 3.1. An important virtue of such rank tests is that the test statistic and its limiting behavior under the null hypothesis are independent of the distribution F generating the observations. The power of the test is obviously dependent on F, but the statistic itself does not require estimation of a nuisance parameter like the sparsity parameter required for the Wald tests or the error variance in the case of least-squares procedures.

3.5.4 Rank Tests Based on Regression Rankscores

How can these ideas be extended to general regression settings when, under the null, a nuisance regression parameter is present? This question was answered by Gutenbrunner and Jurečková (1992), who observed that the Hájek–Šidák rankscores may be viewed as a special case of a more general formulation for the linear model in which the functions $\hat{a}_{ni}(\tau)$ are defined as solutions of the linear program

$$\max\{y'a | X^\top a = (1-\tau)X^\top 1_n,\ a \in [0,1]^n\}, \tag{3.12}$$

where $[0, 1]^n$ is the n-dimensional unit cube and 1_n is an n-vector of ones. This problem is formally dual to linear program (1.15) defining the regression quantiles.

Whereas the primal quantile regression problem may be viewed as generating the sample quantiles, the dual problem may be seen to generate the order statistics or perhaps more precisely the *ranks* of the observations. What does the solution $\hat{a}(\tau)$ look like for the simplest dual problem? Clearly, at $\tau = 0$, feasibility requires that all of the $\hat{a}_i(0) = 1$, and similarly at $\tau = 1$, $\hat{a}_i(1) = 0$ for all $i = 1, \ldots, n$. Starting at $\tau = 0$, consider increasing τ. How should we modify the $\hat{a}_i$s? Initially we should focus on $y_{(1)} = \min\{y_1, \ldots, y_n\}$ because decreasing its weight has the least impact on the sum $y'a$. Thus, if $y_{(1)} = y_j$, then as τ increases a_j must decrease linearly to satisfy the equality constraint. This is fine until τ reaches $1/n$, but at this point a_j has been driven to 0 and it is allowed to go no further. Now $y_{(2)}$, being the smallest available response, is gradually downweighted, and the process continues until all the observations have achieved weight $a_i = 0$, at which point $\tau = 1$.

Thus, as we observed in Section 1.3, when X is simply an n-vector of ones, the Hájek rankscores $\hat{a}_i(\tau)$ take the value one for $\tau \leq (R_i - 1)/n$, where R_i is the rank of y_i in the sample $y_1, \ldots, y_n$ and $\hat{a}_i(\tau) = 0$ for $\tau > R_i/n$. In the interval $[(R_i - 1)/n, R_i/n)$, $\hat{a}_i(\tau) = R_i/n - \tau$, and so we have linear interpolation. More generally, in regression the rankscore functions need not be monotone decreasing, but $\hat{a}_i(\tau) = 1$ when (x_i, y_i) is below the fitted plane, $H(x) = x^\top \hat{\beta}(\tau)$, and $\hat{a}_i(\tau) = 0$ when $y_i > x_i^\top \hat{\beta}(\tau)$. For $\tau \in (0, 1)$ such that $y_i = x_i^\top \hat{\beta}(\tau)$, $\hat{a}_i(\tau)$ lies somewhere between zero and one.

In Figure 3.3 we illustrate the regression rankscores for the well-known stackloss data of Brownlee (1965). There are 21 daily observations on a plant designed for the oxidation of ammonia to produce nitric acid. There are three covariates and so four parameters are being estimated at each quantile. A curious feature of the data is that 8 of the 21 observations, numbers $\{6, 7, 13, 14, 16, 17, 18, 19\}$, have zero residuals at the $\tau = 0.25$ fit. The "rank" given above each subplot is the Wilcoxon "score," that is, the integral of the regression rankscore function with respect to Lebesgue measure.

Asymptotically, the quantile regression rankscore process, $\hat{a}_n(\tau)$, behaves like a weighted empirical process based on the "true" rankscores:

$$\tilde{a}_{ni}(\tau) = I\left(y_i > x_i^\top \beta(\tau)\right).$$

To explore this further, we consider the location-scale linear quantile regression model,

$$Q_{Y_i}(\tau | x_i) = x_i^\top \beta + \left(x_i^\top \gamma\right) F_u^{-1}(\tau),$$

arising from the iid error model

$$Y_i = x_i^\top \beta + \left(x_i^\top \gamma\right) u_i$$

with $\{u_i\}$ iid from distribution function F_u. Following Gutenbrunner and Jurečková (1992), we will denote the diagonal matrix with entries $x_i^\top \gamma$ by Γ_n and introduce a triangular array $\{v_{ni} : i = 1, \ldots, n\}$ of q-dimensional vectors. The residuals from the *weighted* least-squares projection of $V_n = (v_{ni})$ into the space spanned by $X_n = (x_{ni})$ will be written as

$$\tilde{V}_n = V_n - X_n \left(X_n^\top \Gamma_n^{-1} X_n\right)^{-1} X_n^\top \Gamma_n^{-1} V_n.$$

Linear rank statistics based on the regression rankscore process are obtained by simply integrating to obtain the "score" vector,

$$\hat{b}_{ni} = -\int_0^1 \varphi(\tau) d\hat{a}_{ni}(\tau),$$

where φ is a function of bounded variation that we will assume is constant outside $\mathcal{T} = [\varepsilon, 1 - \varepsilon]$ for some $\varepsilon \in (0, 1/2)$. Asymptotically, these rank statistics behave exactly as the corresponding statistics based on the original Hájek rankscores.

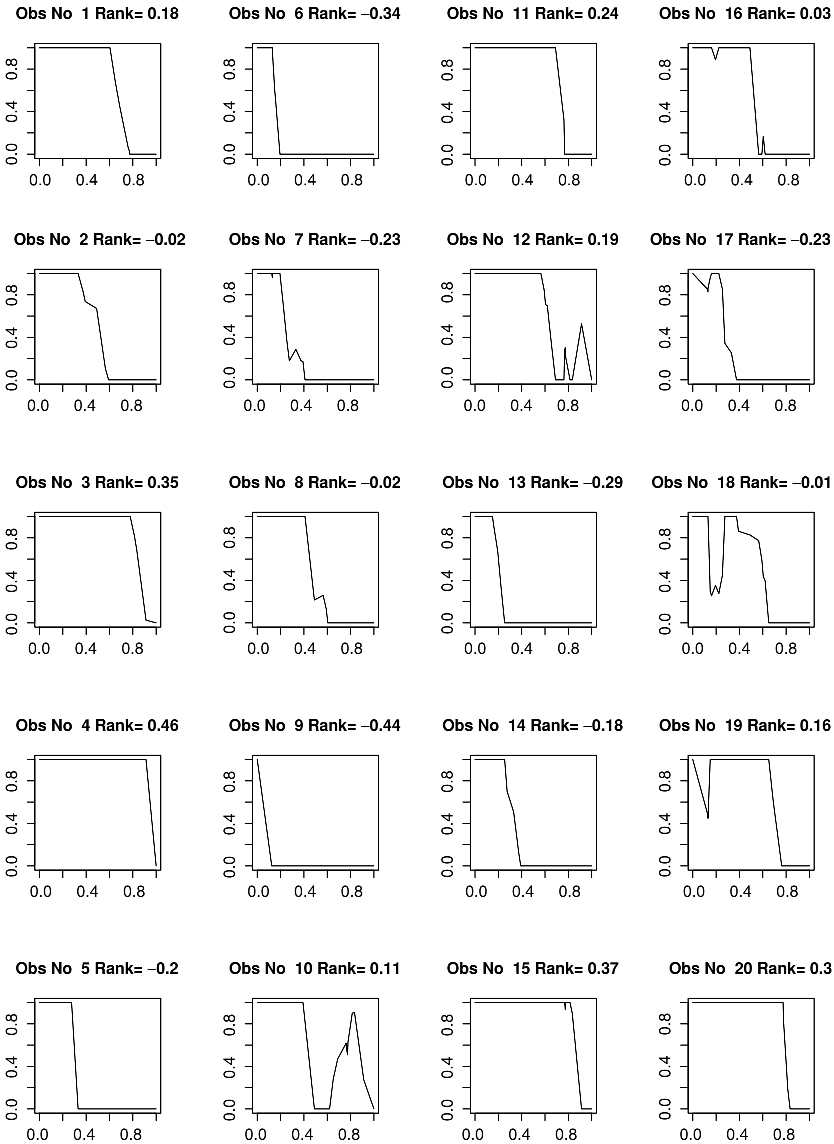

Figure 3.3. Regression rankscores for the stackloss data. For each of the 21 observations, the regression rankscore plots indicate the region of the unit interval for which response lies above the fitted quantile regression hyperplane ($\hat{a}_i(\tau) = 1$), below the fitted hyperplane ($\hat{a}_i(\tau) = 0$), and the region in which the response lies on the fitted hyperplane ($\hat{a}_i(\tau) \in (0, 1)$). Integrating the rankscore functions with respect to Lebesgue measure and subtracting $1/2$ yields the normalized "ranks" indicated above each panel of the plot.

For the classical Wilcoxon score function $\varphi(\tau) = \tau - 1/2$ centered at zero, integration by parts yields

$$\begin{aligned}\hat{b}_{ni} &= -\int_0^1 (s - 1/2) d\hat{a}_{ni}(s) \\ &= \int_0^1 \hat{a}_{ni}(s) ds - 1/2 \\ &= \sum_{j=1}^{J} \frac{1}{2}(\hat{a}_{ni}(\tau_{j+1}) - \hat{a}_{ni}(\tau_j)) - 1/2,\end{aligned}$$

where the last sum is over the J breakpoints $\{\tau_j\}$ of the quantile regression process. As we see, this is just a recentered form of the rescaled ranks introduced earlier.

Regression rankscore tests play the role of Lagrange multiplier, or score, tests in quantile regression. Consider the model

$$Q_{Y_i}(\tau|x_i, z_i) = x_i^\top \beta(\tau) + z_i^\top \zeta(\tau)$$

with the associated null hypothesis,

$$H_0 : \zeta(\tau) = 0,$$

and local alternatives,

$$H_n : \zeta_n(\tau) = \zeta_0(\tau)/\sqrt{n}.$$

Estimating the restricted form of the model, we obtain

$$\hat{a}_n(\tau) = \text{argmax}\{y'a | X^\top a = (1 - \tau)X^\top 1_n,\ a \in [0, 1]^n\},$$

from which we can compute the test statistic,

$$T_n = S_n^\top M_n^{-1} S_n / A^2(\varphi), \tag{3.13}$$

where

$$\begin{aligned} S_n &= n^{-1/2}(Z - \hat{Z})^\top \hat{b}_n \\ \hat{Z} &= X(X^\top \Psi X)^{-1} X^\top \Psi Z \\ \Psi &= \text{diag}(f_i(Q_{Y_i}(\tau|x_i, z_i))) \\ M_n &= (Z - \hat{Z})(Z - \hat{Z})^\top / n \\ \hat{b}_n &= \int_0^1 \hat{a}_n(s) d\varphi(s). \end{aligned}$$

Koenker and Machado (1999) show under mild conditions on the score function φ that T_n has a limiting central χ_q^2 theory under the null, where q denotes the dimension of the parameter ζ. Under the local alternatives, T_n has a noncentral χ_q^2 theory with noncentrality parameter, $\eta(\varphi, \zeta) = \xi^\top M_n^{-1} \xi / A^2(\varphi)$, and $\eta(\varphi, \zeta, F) = \int_0^1 \zeta(s) f(F^{-1}(s)) d\varphi(s)$. Thus, T_n can be used either to explore a global effect of the covariates, Z, on the response Y, or by choosing the φ

to focus exclusively on a single τ as a purely local, quantile-specific test. We describe the latter strategy in more detail in the next subsection.

The choice of the score function φ should obviously be tailored to the plausible forms of the alternatives $\zeta_0(\cdot)$. In the simplest case that $\zeta_0(t) \equiv \zeta_0$, a fixed q-vector, z_i exerts a pure location shift effect; there is a large literature on classical rank tests for location shift to guide this selection. In this case, power is optimized as we have seen by choosing

$$\varphi(s) = \frac{-f'(F^{-1}(s))}{f(F^{-1}(s))}$$

so that $\int f(F^{-1}(s))d\varphi(s) = \int \varphi^2 ds = \mathcal{I}(F)$, the Fisher information for location of the innovation distribution. Similarly, we can consider tailoring the score function to the hypothesis that the z_is exert a pure scale shift. In either case, fully efficient tests can be designed under the (somewhat implausible) preconception that the innovation density is known.

There is a natural tendency to think that such rank tests sacrifice efficiency in comparison to conventional likelihood-based testing methods; however, as we have noted, this need *not* be the case. Indeed, it is worth recalling in this context the celebrated Chernoff–Savage result that the normal scores two-sample rank test of location shift has the same asymptotic power as the two-sample Student t-test at the Gaussian model and strictly *better* asymptotic power than the t-test for local alternatives with non-Gaussian innovations. This result generalizes to efficiency comparisons of the T_n tests described earlier with the classical regression F test in the location shift mode (see, e.g., Koenker, 1996; Hallin and Jurečková, 1999).

3.5.5 Confidence Intervals Based on Regression Rankscores

An important application of the foregoing general theory of rank tests involves the construction of confidence intervals for individual parameters of the quantile regression model. The score function

$$\varphi_\tau(t) = \tau - I(t < \tau)$$

enables us to focus the attention of the test entirely at the τth quantile. The median case

$$\varphi_{1/2}(t) = \frac{1}{2}\,\mathrm{sgn}\,\left(t - \frac{1}{2}\right)$$

is an important special case that is usually referred to as sign-scores.

Specializing to the scalar ζ case and using the τ-quantile score function

$$\varphi_\tau(t) = \tau - I(t < \tau)$$

and proceeding as before, we find that

$$\hat{b}_{ni} = -\int_0^1 \varphi_\tau(t) d\hat{a}_{ni}(t) = \hat{a}_{ni}(\tau) - (1 - \tau) \tag{3.14}$$

with $\bar{\varphi} = \int_0^1 \varphi_\tau(t)dt = 0$ and $A^2(\varphi_\tau) = \int_0^1 (\varphi_\tau(t) - \bar{\varphi})^2 dt = \tau(1-\tau)$. A test of the hypothesis $H_0 : \zeta = \zeta_0$ may be based on $\hat{a}_n$ from solving

$$\max\{(y - z\zeta_0)^\top a | X^\top a = (1-\tau)X^\top 1, a \in [0,1]^n\} \tag{3.15}$$

and the fact that

$$S_n(\zeta_0) = n^{-1/2} x_2^\top \hat{b}_n(\zeta_0) \rightsquigarrow \mathcal{N}\left(0, A^2(\varphi_\tau) q_n^2\right) \tag{3.16}$$

under H_0, where $q_n^2 = n^{-1} z^\top (I - X(X^\top X)^{-1} X^\top) z$. That is, we may compute

$$T_n(\zeta_0) = S_n(\zeta_0)/(A(\varphi_\tau) q_n)$$

and reject H_0 if $|T_n(\zeta_0)| > \Phi^{-1}(1 - \alpha/2)$.

If we would like a confidence interval for ζ, we can invert this rank test, that is, find the values ζ_0 such that the test fails to reject. This takes us back to linear program (3.15), which may now be viewed as a one-parameter parametric linear programming problem in ζ. In ζ the dual vector $\hat{a}_n(\zeta)$ is piecewise constant; ζ may be altered without compromising the optimality of $\hat{a}_n(\zeta)$ as long as the signs of the residuals in the primal quantile regression problem do not change. When ζ gets to such a boundary, the solution does change, but optimality may be restored by taking one simplex pivot. The process may continue in this way until $T_n(\zeta)$ exceeds the specified critical value. Since $T_n(\zeta)$ is piecewise constant, linearly interpolating in ζ may be used to obtain the desired level for the confidence interval. See Beran and Hall (1993) for a detailed analysis of the salutary effect of interpolation like this in the case of confidence intervals for ordinary quantiles. This interval, unlike the Wald-type sparsity intervals, is not symmetric, but it *is* centered on the point estimate $\hat{\beta}_2(\tau)$ in the sense that $T_n(\hat{\beta}_2(\tau)) = 0$. This follows immediately from the constraint $X^\top \hat{a} = (1-\tau)X^\top 1$ in the full problem.

The primary virtue of this approach is that it inherits the scale invariance of the test statistic T_n and therefore circumvents the problem of estimating the sparsity function. Implemented in R, using an adaptation of the algorithm described by Koenker and d'Orey (1993), it has essentially the same computational efficiency as the sparsity methods. A more detailed description of the computational aspects of the parametric programming problem is provided in Section 6.3.

3.6 QUANTILE LIKELIHOOD RATIO TESTS

Having now introduced variants of the Wald and score test for quantile regression, it is natural to investigate analogs of the likelihood ratio test as well. Koenker and Bassett (1982b) have shown that for median regression a test of the hypothesis

$$H_0 : R\beta = r \tag{3.17}$$

in the iid-error linear model,

$$y_i = x_i^\top \beta + u_i, \tag{3.18}$$

could be based on the statistic

$$L_n = 8\left(\tilde{V}\left(\frac{1}{2}\right) - \hat{V}\left(\frac{1}{2}\right)\right)/s\left(\frac{1}{2}\right), \tag{3.19}$$

where we will denote the value of the objective function at the unrestricted minimizer $\hat{\beta}(\frac{1}{2})$ by

$$\hat{V}(\tau) = \min_{\{b \in \mathbb{R}^p\}} \sum \rho_\tau \left(y_i - x_i^\top b\right) \tag{3.20}$$

and

$$\tilde{V}(\tau) = \min_{\{b \in \mathbb{R}^p | Rb = r\}} \sum \rho_\tau \left(y_i - x_i^\top b\right) \tag{3.21}$$

denotes the value under the restricted estimator $\tilde{\beta}(\tau)$. It was shown that under H_0, T_n is asymptotically χ_q^2 where $q = \text{rank}(R)$. That this statistic is related to a likelihood ratio is easy to see. Consider the standard Laplace (double exponential) density,

$$f(u) = \frac{1}{4}\exp\left(-\frac{1}{2}|u|\right).$$

Under the assumption that $\{u_i\}$ in (3.18) comes from this density, we have the log-likelihood

$$\ell(\beta) = -n\log(4) - \frac{1}{2}\sum_{i=1}^{n}\left|y_i - x_i^\top \beta\right|,$$

and the likelihood ratio statistic for testing H_0 would be

$$2(\ell(\hat{\beta}) - \ell(\tilde{\beta})) = 2(\tilde{V}(1/2) - \hat{V}(1/2)),$$

which we would expect to have a χ_q^2 asymptotic distribution under H_0. How does this relate to statistic (3.19)? Where does the factor 8 come from? Note that in the standard Laplace case $s(1/2) = (f(0))^{-1} = 4$, and so the usual theory of the likelihood ratio is vindicated – for this very special case, twice the log-likelihood ratio converges in distribution to χ_q^2 under H_0. However, when the standard Laplace assumption fails to hold, $s(1/2)$ may be quite different from 4 and the likelihood ratio statistic needs to be modified accordingly.

The simplest example of this, an example that yields another variant of the likelihood ratio, is the Laplace density with free scale parameter σ. Now,

$$f(u) = \frac{1}{4\sigma}\exp\left\{-\frac{1}{2\sigma}|u|\right\},$$

and so the log-likelihood is

$$\ell(\beta, \sigma) = -n \log(4\sigma) - \frac{1}{2\sigma} \sum \left| y_i - x_i^\top \beta \right|.$$

It is easily seen that the maximum likelihood estimator of σ is

$$\hat{\sigma} = n^{-1} \hat{V}\left(\frac{1}{2}\right) = \frac{1}{2n} \sum \left| y_i - x_i^\top \hat{\beta} \right|$$

for the unrestricted model and, similarly, $\tilde{\sigma} = n^{-1} \tilde{V}(\frac{1}{2})$ is the maximum likelihood estimator under H_0. Concentrating the likelihood with respect to β we have

$$\ell(\hat{\beta}, \hat{\sigma}) = -n \log \hat{\sigma} - n - n \log 4,$$

and with an analogous definition of $\ell(\tilde{\beta}, \tilde{\sigma})$ the likelihood ratio statistic becomes

$$2(\ell(\hat{\beta}, \hat{\sigma}) - \ell(\tilde{\beta}, \tilde{\sigma})) = 2n \log(\tilde{\sigma}/\hat{\sigma}).$$

Again, we are entitled to expect that the likelihood ratio statistic will have a limiting χ_q^2 behavior. This can be easily checked using the earlier result by writing

$$\log(\tilde{\sigma}/\hat{\sigma}) = \log(1 + (\tilde{\sigma} - \hat{\sigma})/\hat{\sigma}) \approx (\tilde{\sigma} - \hat{\sigma})/\hat{\sigma}, \tag{3.22}$$

an approximation whose validity is easy to establish under the null. Dividing numerator and denominator by $\sigma \equiv E|u|$, and arguing that $\hat{\sigma}/\sigma \to 1$, we have

$$\begin{aligned} 2n \log(\tilde{\sigma}/\hat{\sigma}) &= 2n(\tilde{\sigma} - \hat{\sigma})/\sigma + o_p(1) \\ &= 2(\tilde{V}(1/2) - \hat{V}(1/2))/\sigma + o_p(1). \end{aligned} \tag{3.23}$$

Noting that $s(1/2) = 4\sigma$ in this case completes the argument.

Although the unknown-scale form of the Laplace model is certainly preferable to the original fixed-scale form, it is still unsatisfactory in that any departure from the condition $s = 4\sigma$ wreaks havoc with the asymptotic behavior of the test statistic under the null. As long as the errors are iid, this is easily rectified by defining the modified likelihood ratio statistics,

$$L_n(1/2) = 8(\tilde{V}(1/2) - \hat{V}(1/2))/s(1/2), \tag{3.24}$$

or again, using (3.22),

$$\Lambda_n(1/2) = 8n\sigma \log(\tilde{\sigma}/\hat{\sigma})/s(1/2), \tag{3.25}$$

which may be shown to be asymptotically equivalent with limiting χ_q^2 behavior. By directly evaluating the effect of the restriction on the mean absolute deviation from the median regression estimate, this result provides a useful complement to alternative Wald and score formulations of tests of H_0. The modified likelihood ratio tests obviously require estimation of the nuisance parameters σ and s, but this is quite straightforward. For σ we may simply use $n^{-1}\hat{V}(1/2)$, whereas for s we may use the sparsity estimation methods described earlier. It should be emphasized, however, that the foregoing theory for the likelihood ratio form of the test requires that the conditional density of the response be constant along the conditional median function. This would be true in the classical iid error model, where the covariates have a pure location-shift effect, but in more general settings one would need to reweight the terms in the objective function to obtain limiting χ^2 behavior.

The same approach we have elaborated for the median may be extended immediately to other quantiles. Define $\hat{V}(\tau)$ and $\tilde{V}(\tau)$ as in Section 3.2, let $\hat{\sigma}(\tau) = n^{-1}\hat{V}(\tau)$, $\tilde{\sigma}(\tau) = n^{-1}\tilde{V}(\tau)$, and for the τth quantile consider the test statistics

$$L_n(\tau) = \frac{2}{\lambda^2(\tau)s(\tau)}[\tilde{V}_n(\tau) - \hat{V}_n(\tau)]$$

and

$$\Lambda_n(\tau) = \frac{2n\hat{\sigma}(\tau)}{\lambda^2(\tau)s(\tau)}\log(\tilde{\sigma}(\tau)/\hat{\sigma}(\tau)),$$

where $\lambda^2(\tau) = \tau(1-\tau)$. Tests of this sort based on the drop in the optimized value of the objective function of an M-estimator when relaxing the restriction imposed by the hypothesis are termed ρ-tests by Ronchetti (1985). Following this terminology, we will refer to tests based on $L_n(\tau)$ and $\Lambda_n(\tau)$ as quantile ρ-tests, although the phrase quasi-likelihood-ratio tests would also be appropriate.

3.7 INFERENCE ON THE QUANTILE REGRESSION PROCESS

We have seen that the regression rankscore process can be employed to construct regression analogs of the well-established rank tests of the location–scale-shift hypotheses developed for the p-sample model. They can also be adapted, using the quantile score function, to test the hypothesis that particular covariates exert no effect at selected quantiles. This ability to focus attention on particular quantiles is especially valuable because inference based on global hypotheses risks averaging out important effects. A finding that a covariate has no location-shift effect is sometimes taken to imply that it has no effect whatsoever. But it is not unusual to find situations in which a covariate exerts significant negative effects in one tail of the conditional distribution while exerting a positive effect in the other tail. These effects may cancel out in an evaluation of the “average” location effect but still be of substantial interest.

Classical goodness-of-fit tests offer a natural way to expand inference for the quantile regression model. Rather than asking, in the two-sample model, if the treatment distribution differs from the control distribution by a location shift, we ask instead if it differs in any way at all. This leads us to variants of the two-sample Kolmogorov–Smirnov and Cramér –von Mises tests. Rather than formulating tests based on the difference in empirical distribution functions, however, we can formulate tests based on differences in the empirical quantile functions. In the simplest two-sample setting we can consider the test statistic

$$T_n = \sup_{\tau \in \mathcal{T}} \left| G_m^{-1}(\tau) - F_n^{-1}(\tau) \right|,$$

where F_n and G_m denote the control and treatment empirical distribution functions, respectively. We will take $\mathcal{T}$ to be a closed subinterval of (0, 1), typically $[\epsilon, 1-\epsilon]$, for some $\epsilon \in (0, 1/2)$. Other forms of the test statistic based on L_p norms, for example,

$$T_n = \int_{\mathcal{T}} \left| G_m^{-1}(\tau) - F_n^{-1}(\tau) \right| d\tau$$

or

$$T_n = \left(\int_{\mathcal{T}} \left(G_m^{-1}(\tau) - F_n^{-1}(\tau) \right)^2 d\tau \right)^{-1/2},$$

may also be considered. The latter L_2 form of the test is closely related to Wasserstein distance; see, for example the discussion by Shorack and Wellner (1986) in their Section 2.6.

As we have seen in Chapter 2, the quantile treatment effect in the two-sample model,

$$\delta(\tau) = G^{-1}(\tau) - F^{-1}(\tau),$$

may be estimated by solving the bivariate quantile regression problem

$$(\hat{\alpha}(\tau), \hat{\delta}(\tau)) = \operatorname{argmin} \sum_{i=1}^{n} \rho_\tau(y_i - \alpha - x_i \delta),$$

where x_i is an indicator of the treatment; that is, $x_i = 1$ if y_i is a treatment response, and $x_i = 0$ if y_i is a control observation.

Under the null hypothesis that the treatment and control distributions are identical, tests based on the large sample theory of the rankscore, Wald, and ρ-processes just introduced can be constructed. Critical values of the tests can be evaluated by simulation of the underlying Brownian bridge approximations. Quantile regression offers a natural extension of these testing principles to the general linear regression setting. Covariates need not be binary, or even discrete. Given the standard model,

$$Q_{y_i}(\tau | x_i) = x_i^\top \beta(\tau), \tag{3.26}$$

we can formulate the null hypothesis,

$$H_0: \; R\beta(\tau) = r(\tau) \qquad \tau \in \mathcal{T}, \tag{3.27}$$

and consider tests of H_0 against the sequence of alternatives:

$$H_n: \; R\beta(\tau) - r(\tau) = \zeta(\tau)/\sqrt{n} \qquad \tau \in \mathcal{T}. \tag{3.28}$$

This might entail, for instance, the assertion that some subset of the covariates exert no influence on the conditional quantiles of the response over a specified range of quantiles, $\tau \in \mathcal{T}$. Such exclusion restrictions may be expressed by setting $R = [0 \vdots I_q]$ and $r(\tau) = 0$ for an appropriate partitioning of the parameter vector.

To investigate the asymptotic behavior of such statistics, we require some rather basic theory and notation regarding Bessel processes. Let $W_q(\tau)$ denote a q-vector of independent Brownian motions and, thus, for $\tau \in [0, 1]$,

$$B_q(t) = W_q(t) - \tau W_q(1)$$

will represent a q-vector of independent Brownian bridges. Note that, for any fixed $\tau \in (0, 1)$,

$$B_q(\tau) \sim \mathcal{N}(0, \tau(1-\tau)I_q).$$

The normalized Euclidean norm of $B_q(\tau)$,

$$\mathcal{B}_q(\tau) = \| \; B_q(\tau) \; \| \; / \sqrt{\tau(1-\tau)},$$

is generally referred to as a Bessel process of order q. Critical values for $\sup \mathcal{B}_q^2(\tau)$ have been tabled by DeLong (1981) and, more extensively, by Andrews (1993) using simulation methods. The seminal work on Bessel processes and their connection to K-sample goodness-of-fit tests is Kiefer's (1959). Again, for any fixed $\tau \in (0, 1)$ we have that $\mathcal{B}_q^2(\tau) \sim \chi_q^2$. Thus, we may interpret $\mathcal{B}_q^2(\tau)$ as a natural extension of the familiar univariate χ^2 random variable with q degrees of freedom. Note that, in the special case $q = 1$, $\sup \mathcal{B}_1^2(\cdot)$ behaves asymptotically like a squared Kolmogorov–Smirnov statistic.

To characterize the behavior of the test statistic under local alternatives, it is helpful to define a noncentral version of the squared Bessel process as an extension of the noncentral χ^2 distribution. Let $\mu(\tau)$ be a fixed, bounded function from $[0, 1]$ to $\mathbb{R}^q$. The standardized squared norm

$$\| \; \mu(\tau) + B_q(\tau) \; \|^2 \; / (\tau(1-\tau))$$

will be referred to as a noncentral Bessel process of order q with noncentrality function $\eta(\tau) = \mu(\tau)^\top \mu(\tau)/(\tau(1-\tau))$ and will be denoted by $\mathcal{B}_{q,\eta(\tau)}^2$. Of course, for any fixed $\tau \in (0, 1)$, $\mathcal{B}_{q,\eta(\tau)}^2 \sim \chi_{q,\eta(\tau)}^2$, a noncentral χ_q^2 random variable with q degrees of freedom and noncentrality parameter $\eta(t)$.

3.7.1 Wald Processes

Under null hypothesis (3.27), the Wald process,

$$\nu_n(\tau) = \sqrt{n}[RV(\tau, \tau)R^\top]^{-1}(R\hat{\beta}(\tau) - r(\tau)),$$

with $V(\tau, \tau)$ as in (3.8), converges weakly to a q-dimensional Brownian bridge. And under the sequence of local alternatives, the statistic

$$\mathcal{B}_n(\tau) = \| \; \nu_n(\tau) \; \| \; / \sqrt{\tau(1-\tau)}$$

behaves asymptotically like a noncentral Bessel process with noncentrality parameter $\eta = \zeta^\top [RV(\tau,\tau)R^\top]^{-1}\zeta/(\tau(1-\tau))$. Variants of this theory are considered in more detail by Koenker and Machado (1999).

3.7.2 Quantile Likelihood Ratio Processes

The quantile likelihood ratio, or ρ-tests, considered earlier can also be adapted to test hypothesis (3.27). However, as has been already emphasized, some reweighting of the terms in the objective function may be necessary to accommodate variation in the conditional density over the design space.

3.7.3 The Regression Rankscore Process Revisited

The regression rankscore process can also be used to formulate tests of null hypothesis (3.27). Reparameterizing, we can reduce the hypothesis to an exclusion restriction. Let P be a matrix such that $A = [P\vdots R]^\top$ is nonsingular and $R^\top P = 0$. Transforming

$$\begin{aligned}
\tilde{y} &= y - XR(R^\top R)^{-1}R^\top r(\tau)\\
\tilde{X} &= XP(P^\top P)^{-1}\\
Z &= XR(R^\top R)^{-1}
\end{aligned}$$

allows us to use test statistic (3.13) with the quantile score function $\varphi_\tau(t) = \tau - I(t < \tau)$. As shown by Koenker and Machado (1999), the resulting rankscore process,

$$T_n(\tau) = S_n^\top(\tau)M_n^{-1}S_n(\tau)/A^2(\varphi_\tau),$$

has the same noncentral Bessel process representation as the Wald process described earlier in this section.

3.8 TESTS OF THE LOCATION-SCALE HYPOTHESIS

At various stages we have considered the linear location–scale model

$$y_i = x_i^\top \alpha + \left(x_i^\top \gamma\right) u_i,$$

where the u_i are assumed to be iid from some distribution F_0. In this case, the conditional quantile functions of the response can be written as

$$Q_{y_i}(\tau|x_i) = x_i^\top \beta(\tau),$$

where $\beta(\tau) = \alpha + \gamma F_0^{-1}(\tau)$. Thus, all p components of the vector function β are – under the location-scale hypothesis – identical up to an affine transformation. Alternatively, we may wish to test the more restrictive hypothesis that we have a pure location-shift model so that $x_i^\top \gamma = 1$. In this section we will describe a general strategy designed to test such hypotheses. In the two-sample treatment control model, these hypotheses have the interpretation that the treatment and control distributions differ either by a location shift or by a location–scale shift. It is natural to consider tests of the Kolmogorov–Smirnov type; however, the introduction of unknown nuisance parameters into the null hypothesis raises some new and rather challenging issues.

The problem posed in this section is closely related to the classical problem of one-sample goodness-of-fit tests based on the empirical distribution function when there are unknown nuisance parameters under the null hypothesis. Rather than testing that observations arose from a *standard* normal distribution, we may wish to test that they arose from *some* normal distribution with unspecified mean and variance parameters. If, proceeding naively, we compute $\hat{\theta}_n = (\hat{\mu}_n, \hat{\sigma}_n)$, then construct the parametric empirical process

$$U_n(y) = \sqrt{n}(F_n(y) - F_{\hat{\theta}_n}(y)),$$

where $F_{\hat{\theta}_n}(y) = \Phi((y - \hat{\mu}_n)/\hat{\sigma}_n)$, and compute the Kolmogorov–Smirnov statistic,

$$K_n = \sup \| U_n(y) \|,$$

close examination of the limiting theory shows that U_n under mild conditions has a limiting Gaussian process, but with covariance function that depends on the estimation of the parameter θ. Consequently, K_n does not have the desired Kolmogorov–Smirnov distribution. There has been considerable discussion of various approaches to deal with this problem, notably the work of Durbin (1973).

Khmaladze (1981) offered the first general approach to these one-sample goodness-of-fit problems with estimated parameters. Fortunately, Khmaladze's general approach extends naturally to the related problems in the quantile regression setting. This has been explored by Koenker and Xiao (2002).

As we have seen, in the location-scale model, the process

$$\sqrt{n}\varphi_0(\tau)\Omega^{-1/2}(\hat{\beta}(\tau) - \beta(\tau))$$

converges on compact sets; $\mathcal{T} \subset (0, 1)$ to a p-dimensional independent Brownian bridge process. Here, $\varphi_0(\tau) = f_0(F_0^{-1}(\tau))$ and $\Omega = H_0^{-1} J_0 H_0^{-1}$ with $J_0 = \lim n^{-1} \sum x_i x_i^\top$ and $H_0 = \lim n^{-1} \sum x_i x_i^\top/(\gamma^\top x_i)$. And so it follows that a test of the hypothesis

$$R\beta(\tau) = r(\tau)$$

can be based on the process

$$v_n(\tau) = \sqrt{n}\varphi_0(t)(R\Omega R^\top)^{-1/2}(R\hat{\beta}(\tau) - r(\tau)),$$

provided that R and r are completely specified. Nuisance parameters involving φ_0 and Ω can be replaced without jeopardizing the first-order asymptotics of the tests. However, the location-scale shift hypothesis is not so straightforward.

Consider the problem of testing the pure location-shift hypothesis that all the slope coefficients of the quantile regression model are constant over $\tau \in \mathcal{T}$. We can write this in $R\beta = r$ form by taking $R = [0 \vdots I_{p-1}]$ and $r = (\alpha_2, \ldots, \alpha_p)^\top$: a fixed, but unknown vector. The natural expedient of replacing the unknown r in the test statistic by estimates introduces some fundamental difficulties. The process

$$\hat{\nu}_n(\tau) = \sqrt{n}\varphi_0(\tau)(R\Omega R^\top)^{-1/2}(R\hat{\beta}(\tau) - \hat{r}(\tau))$$

remains asymptotically Gaussian; however, the estimation of r introduces a new drift component that complicates the covariance kernel of the process and disrupts the asymptotically distribution-free character of the limiting theory. One way to resolve this difficulty, described in more detail in the next chapter, employs the Khmaladze (1981) martingale transformation. In effect, this involves replacing $\hat{\nu}(t)$ by the residual process obtained by a continuous time recursive least-squares regression of $\hat{\nu}_n(t)$ on the score functions defining the estimator of $\hat{r}$. After transformation, we have

$$\tilde{\nu}_n = Q_g \hat{\nu}_n \Rightarrow w_0,$$

a standard Brownian motion, with $p - 1$ independent coordinates. Tests based on the Kolmogorov–Smirnov-type statistic

$$K_n = \sup_{\tau \in \mathcal{T}} \|\tilde{\nu}_n(\tau)\|$$

then have easily simulated critical values.

The choice of the norm used in the definition of K_n is obviously also an issue. Koenker and Xiao (2002) suggested that the ℓ_1-norm avoids accentuating the effect of one or two extreme coordinates and therefore may be preferable to the usual Euclidean norm. The index set $\mathcal{T}$ is typically chosen to be an interval of the form $[\epsilon, 1 - \epsilon]$, but may be chosen to focus attention on one tail or some other feature of the distribution if this better suits the application. In some situations it is desirable to restrict the interval of estimation to a closed subinterval $[\tau_0, \tau_1]$ of $(0, 1)$. This can be easily accommodated by considering the renormalized test statistic

$$K_n = \sup_{\tau \in \mathcal{T}} \|\tilde{\nu}_n(\tau) - \tilde{\nu}_n(\tau_0)\| / \sqrt{\tau_1 - \tau_0}.$$

Koenker and Xiao (2002) illustrate the role of the foregoing tests in an application to model the effect of unemployment insurance rules on the duration of spells of unemployment. The data are taken from the Pennsylvania unemployment bonus experiments conducted in 1988–89. In this period, new claimants for unemployment insurance were randomized into one of several treatment groups or a control group. Control participants abided by the usual rules of the unemployment insurance system; treatment participants were offered a cash bonus to be awarded if the claimant was certifiably reemployed within a specified qualification period. For simplicity we will focus on only one of the treatment groups: those offered a bonus of six times their weekly benefit provided reemployment was established within 12 weeks. For this group the bonus averaged about $1000 for those collecting it. For a more detailed analysis incorporating the other treatments, see Koenker and Bilias (2001).

The model proposed to analyze the experimental data is the quantile regression model of the well-known accelerated failure time (AFT) model. Conditional quantiles of the logarithm of the unemployment durations (in weeks) are assumed to take the linear form

$$Q_{\log(T)}(\tau, x) = x^\top \beta(\tau).$$

There is an extensive set of covariates associated with each participant, but these were reduced to the following:

- Indicator for the treatment group.
- Indicators for female, black, and Hispanic respondents.
- Number of dependents, with 2 indicating two or more dependents.
- Indicators for the five quarters of entry to the experiment.
- Indicator for whether the claimant "expected to be recalled" to a previous job.
- Indicators for whether the respondent was "young" – less than 35 – or "old" – indicating age greater than 54.
- Indicator for whether claimant was employed in the durable goods industry.
- Indicator for whether the claimant was registered in one of the low unemployment short-unemployment-duration districts: Coatesville, Reading, or Lancaster.

The choice of the log transformation was dictated primarily by the desire to achieve linearity of the parametric specification and its convenience of interpretation. When the covariate effects influence only the location of the response $\log(T)$, we can write the restricted model as

$$\log(T_i) = x_i^\top \beta + u_i$$

with u_i iid, yielding the classical form of the AFT. Furthermore, if the distribution function of the errors takes the particular form $F(u) = 1 - e^{-e^u}$, then we have the Cox proportional hazard model with Weibull baseline hazard. It is natural to ask whether it is possible to formulate tests of these restrictions.

In Figure 3.4 we present the results from estimating model. We restrict the estimation to the interval $[0.20,\ 0.80]$ because roughly 20% of the sample are immediately reemployed within the first week, and another 20% of the sample do not find work within the 26-week follow-up window of the experiment. These proportions depend to some degree on the nontreatment covariates of the model, but they are not dependent on the treatment, and so little is lost by treating them as fixed. The intercept plot in the upper left can be interpreted as the estimated quantiles of unemployment duration (in log weeks) for a white, middle-aged male entering the experiment in the first quarter, not expected to be recalled to his prior job, not previously employed in the durable

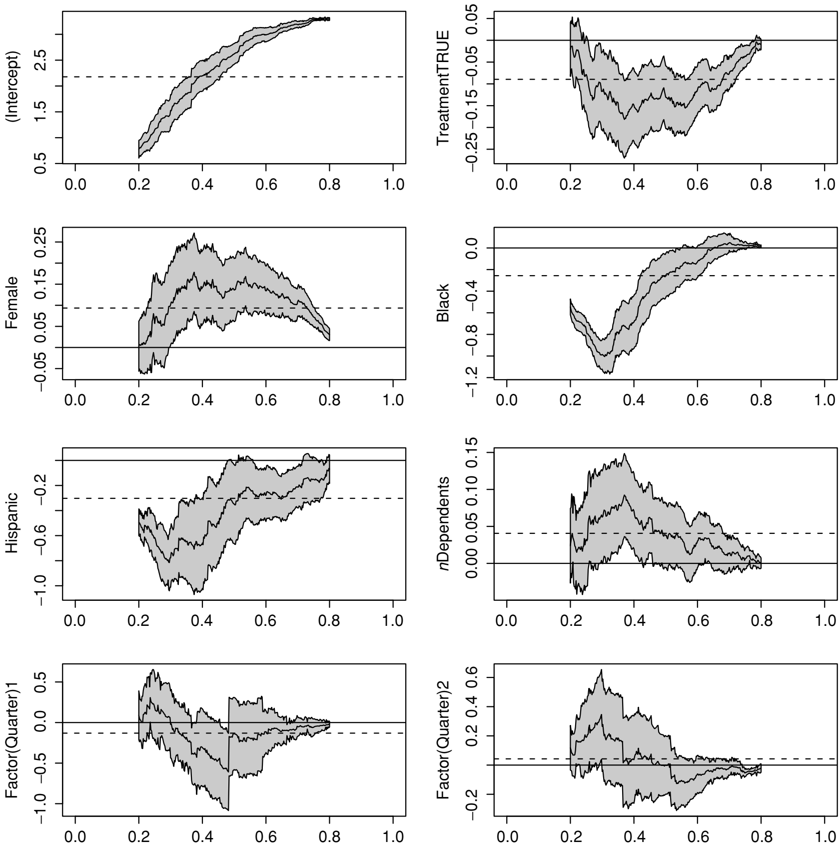

Figure 3.4. Quantile regression process for log duration model. A 90% pointwise confidence band for the quantile regression parameters is indicated by the shaded region. The solid horizontal line indicates $\beta = 0$, the null effect. The least-squares estimate is represented by the dashed line in each of the panels.

goods sector, and not from one of the low-unemployment counties. The treatment effect is not at all what we should expect from a location-shift model: there is essentially no treatment effect near the 0.2 and 0.8 cutoffs, but a rather substantial, almost 15% reduction in durations in the central region of the distribution. This is certainly as we might have expected to see, because the bonus should not have much effect on the extreme outcomes where the bonus is not at issue.

Glossing briefly over the other covariate effects, we find that women and older workers are generally slower to return to work than their younger male counterparts. Blacks and Hispanics also have significantly shorter durations than whites throughout most of the distribution. An expectation to be recalled to a previous job seems to exert a strong disincentive to return to work quickly, but

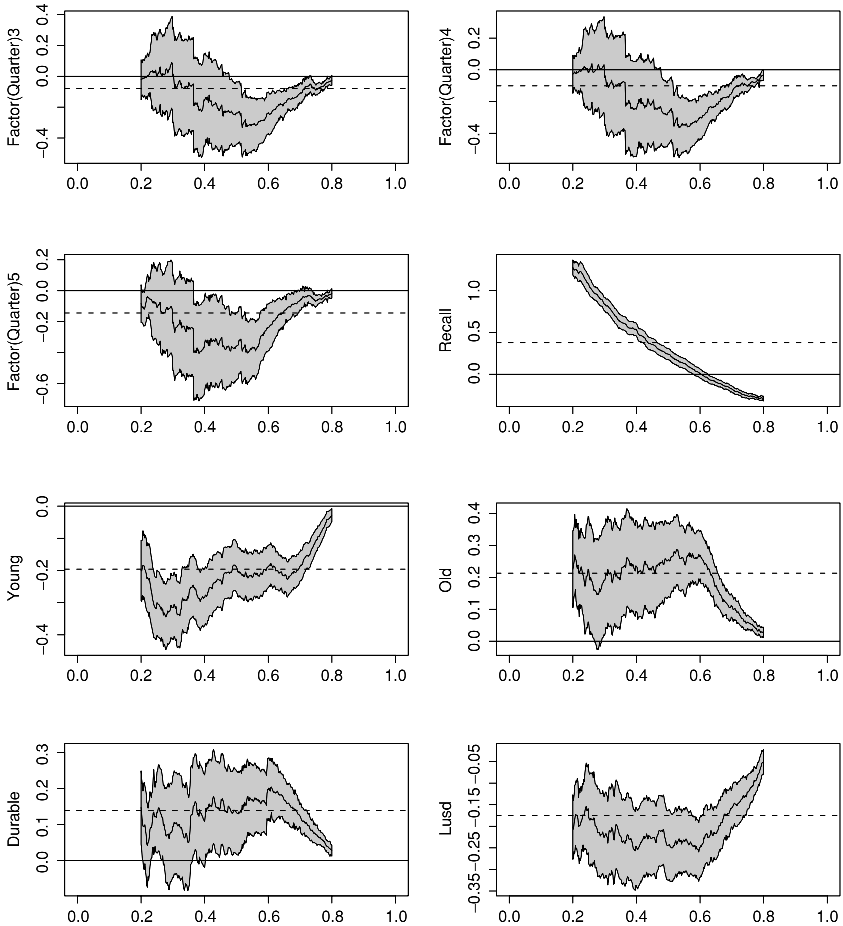

Figure 3.4. (*Continued*)

in the upper tail this reverses itself and there are a significant shorter durations above the 0.6 quantile.

Our tests of the location-and location–scale-shift hypotheses are based on a discretization of the quantile regression process to 301 equally spaced points on the interval [0.20, 0.80]. Recall that if the location-shift hypothesis holds then we would expect to see all the covariate plots in Figure 3.4 looking like random variation around a constant level. Whereas, if the location-scale shift model is appropriate, all the plots should appear as affine shifts of the intercept plot, and so all the coefficients should have roughly the same shape viewed as functions of τ. Neither of these visual impressions appears to be very promising based on Figure 3.4, but we may wish to have a more formal evaluation.

We will try to briefly describe the steps involved in computing the relevant test statistics. We begin by computing the estimates of the nuisance parameters: these are obtained for the location case by simply averaging over the set $\mathcal{T}$, and for the location–scale case by regressing the slope components $\hat{\beta}_i(t), i = 2, \ldots, p$, on $\hat{\beta}_1(t)$ over the 301 "observations" in the set $\mathcal{T}$. We then compute the difference between each coefficient and its prediction under the hypothesis and standardize the resulting $(p-1)$-dimensional process to obtain $\hat{v}_n(t)$. To perform the transformation

$$\tilde{v}_n = Q_g \hat{v}_n,$$

we need to estimate the score functions $\dot{g} = (1, \dot{f}/f, 1 + F^{-1}\dot{f}/f)$. There is a moderately large literature dealing with the estimation of score functions. Koenker and Xiao (2002), following Portnoy and Koenker (1989), employ a variant of Silverman's (1986) adaptive kernel density estimator for the separate numerator and denominator components. Given the estimate of $\dot{g}$ based on the first (intercept) coordinate $\hat{\beta}_1$, the martingale transformation is just a recursive least-squares procedure whose residuals constitute the process $\tilde{v}_n$. At this point we can compute the test statistic K_n.

In this application we take $\mathcal{T} = [0.25,\ 0.75]$ so that there is an additional 5% trimming designed to mollify the erratic behavior of the transformed process in the tails where the recursive estimation is least reliable. The norm is chosen to be $\|a\| = \sum |a_i|$, and we take $\tau_0 = 0.20$ and $\tau_1 = 0.80$. There are 15 slope coefficients under test, and so according to Table B.2 the 1% critical value for the test statistics is 16.00. Table 3.2 reports test statistics of 122.44 and 380.01 for the location–scale- and location-shift hypotheses, respectively. In both cases the superficial implausibility of the hypotheses suggested by the first plot is strongly confirmed by the formal test. Note that these values for the joint test differ slightly from those appearing in the original paper; the tests on the individual coefficients are unchanged.

It is of some independent interest to investigate which of the coordinates contribute "most" to the joint significance of the K_n statistic. This inquiry is fraught with all the usual objections to multiple testing because the coordinates are not independent; but we plunge ahead, nevertheless. In place of the joint hypothesis, we can consider univariate subhypotheses of the form

$$\beta_i(\tau) = \mu_i + \sigma_i \beta_1(\tau)$$

for each "slope" coefficient. In effect, this approach replaces the matrix standardization used for the joint test by a scalar standardization. The martingale transformation is then applied just as in the previous case. Table 3.2 reports the test statistics,

$$K_{ni} = \sup_{\tau \in \mathcal{T}} |\tilde{v}_{ni}(\tau) - \tilde{v}_{ni}(\tau_0)| / \sqrt{\tau_1 - \tau_0},$$

for each of the covariates. Effects for the five quarters of entry are not reported. The critical values for these coordinatewise tests are 1.923 at 0.05 and 2.420 at 0.01, and so the treatment, race, gender, and age effects are highly significant.

Table 3.2. *Tests of the location–scale-shift hypothesis*

Variable	Location–scale shift	Location shift
Treatment	5.41	5.48
Female	4.47	4.42
Black	5.77	22.00
Hispanic	2.74	2.00
N-Dependents	2.47	2.83
Recall effect	4.45	16.84
Young effect	3.42	3.90
Old effect	6.81	7.52
Durable effect	3.07	2.83
Lusd effect	3.09	3.05
Joint effect	122.44	380.01

Also reported in Table 3.2 are the corresponding test statistics for the pure location-shift hypothesis. Not surprisingly, we find that the more restrictive hypothesis of constant $\beta_i(\tau)$ effects is considerably less plausible than the location–scale hypothesis. The joint test statistic is now 380.01, with 0.01 critical value of 16.00, and all of the reported covariate effects are significant at level 0.05, with the exception of the Hispanic effect.

3.9 RESAMPLING METHODS AND THE BOOTSTRAP

Assessing the accuracy of the sample quantiles was one of the earliest success stories for the bootstrap. As noted by Efron (1982), the classical delete-one jacknife estimate of the sampling variance fails for sample quantiles. Suppose $\hat{\theta}_n$ is the median of the sample $\{X_1, \ldots, X_n\}$ and let $\hat{\theta}_{(i)}$ denote the median with the ith observation deleted; then the jacknife estimate of the variance of the median is

$$v_n = \frac{n-1}{n} \sum_{i=1}^{n} \left(\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)}\right)^2,$$

with $\hat{\theta}_{(\cdot)} = n^{-1} \sum \hat{\theta}_{(i)}$. As we have seen, the asymptotic variance of $\sqrt{n}(\hat{\theta}_n - F^{-1}(1/2))$ should equal $1/(2f^2(F^{-1}(1/2)))$; but for $n = 2m$,

$$v_n = \frac{n-1}{4}(x_{m+1} - x_m)^2,$$

and, thus,

$$nv_n \rightsquigarrow \frac{1}{4f^2(F^{-1}(1/2))}\left(\frac{\chi_2^2}{2}\right)^2,$$

where $(\chi_2^2/2)^2$ is a random variable with mean 2 and variance 20. Thus, v_n markedly overstates the sampling variability of the median "on average" and fails to converge to the correct asymptotic variance. In such cases, the delete-one distributions are too close to the empirical distribution of the original sample; the bootstrap rectifies this. By resampling from the empirical, with replacement, the bootstrap distributions differ from the original empirical by $\mathcal{O}(1/\sqrt{n})$, whereas the delete-one distributions differ only by $\mathcal{O}(1/n)$. In this case, this leads to a difference quotient estimate of the density that has too narrow a bandwidth, thus exaggerating its variability.

There is quite an extensive literature on the use of the bootstrap and related resampling schemes for quantile regression. Efron (1982, pp. 35–6) suggested the residual bootstrap for a nonlinear median regression problem, and this form of the bootstrap has been considered by De Angelis, Hall, and Young (1993a), Buchinsky (1994), Hahn (1995), and several others. In the linear quantile regression setting, let

$$\hat{\beta}(\tau) = \text{argmin} \sum \rho_\tau \left(y_i - x_i^\top b \right)$$

and let $\hat{F}_n$ denote the empirical distribution of the residuals, $\hat{u}_i = y_i - x_i \hat{\beta}(\tau)$, with respect to this fit:

$$\hat{F}_n(u) = n^{-1} \sum_{i=1}^{n} I(\hat{u}_i < u).$$

Now, consider drawing bootstrap samples $u_i^*, \ldots, u_n^*$ from $\hat{F}_n(u)$; with replacement, set $y_i^* = x_i \hat{\beta}(\tau) + u_i^*$ and compute

$$\beta_n^*(\tau) = \text{argmin} \sum \rho_\tau \left(y_i^* - x_i^\top b \right).$$

DeAngelis *et al.* show that under iid error conditions the distribution of the bootstrapped β_n^*s,

$$\hat{G}(z) = P(\sqrt{n}(\beta_{nj}^*(\tau) - \hat{\beta}_{nj}(\tau)) \leq z_j,\ \ j = 1, \ldots, p|\mathcal{X}),$$

conditional on the initial sample $\mathcal{X} = \{(x_i, Y_i) : i = 1, \ldots, n\}$ converges to the limiting distribution of $\sqrt{n}(\hat{\beta}_n(\tau) - \beta(\tau))$ and, furthermore, that the error of this approximation is of order $O(n^{-1/4})$ as $n \to \infty$.

To contrast this with the conventional strategy of directly estimating the asymptotic covariance matrix $\omega^2 D_0$, where $\omega^2 = \tau(1-\tau)/f^2(F^{-1}(\tau))$ and $D_0 = \lim n^{-1} \sum x_i x_i^\top$, they observe that $f(F^{-1}(\tau))$ can be estimated nonparametrically to $O(n^{-2/5})$ accuracy. Consequently, the error of approximation of the $\mathcal{N}(0, \hat{\omega}^2 D_0)$ approximation is also of order $O(n^{-2/5})$. This leads to the question of whether the residual bootstrap approximation can be improved by some further smoothing. DeAngelis *et al.* show that sampling from a smoothed version of $\hat{F}_n$ can yield improvement to $O(n^{-2/5})$ for the bootstrap. Knight (2002) provides an explicit form for the distribution of the $n^{1/4}$ normalized discrepancy between $\sqrt{n}(\beta_n^*(\tau) - \hat{\beta}_n(\tau))$ and the bootstrap form of its Bahadur representation.

In applications, we are rarely confident about the iid error, location-shift model and so the residual bootstrap is of limited practical interest for quantile regression. Fortunately, the (x, y) pair form of the bootstrap provides a simple, effective alternative for independent but not identically distributed settings. Instead of drawing bootstrap samples from the empirical distribution of the residuals as we have just described, we may draw samples of the (x_i, y_i) pairs from the joint empirical distribution of the sample. That is, (x_i^*, y_i^*) is drawn *with replacement* from the n pairs $\{(x_i, y_i) : i = 1, \dots, n\}$ of the original sample, each with probability $1/n$. This form of the bootstrap has been widely used in applications of quantile regression.

Given the bootstrap realizations $\{\hat{\beta}_b^*(\tau) : \ b = 1, \dots, B\}$, there are several options that have been considered for constructing tests and confidence intervals. Most straightforwardly, one can compute the empirical covariance matrix of the realizations and construct tests and confidence intervals directly from it. Alternatively, one can form percentile intervals as discussed, for example, by Efron and Tibshirani (1993). Many important practical aspects of the implementation of the bootstrap, including the crucial question of how to choose the number of replications, B, are treated by Andrews and Buchinsky (2000, 2001).

3.9.1 Bootstrap Refinements, Smoothing, and Subsampling

There have been several proposals for refinements of the (x, y) bootstrap based on smoothing. Horowitz (1998) considers smoothing the quantile regression objective function. Under sufficiently smooth conditional density conditions, using higher order kernels, he obtains refinements in the asymptotic accuracy of rejection probabilities based on bootstrap critical values for symmetrical t and χ^2 tests. For univariate quantiles, similar results for smoothed bootstrap procedures are developed by Janas (1993). Horowitz reports simulations showing that conventional $O(n^{-1/5})$ bandwidth rules for sparsity estimation yield too-conservative tests and confidence intervals for the unsmoothed median regression estimator, whereas his smoothed median regression approach appears somewhat less sensitive to bandwidth choice. It would be valuable to have further comparisons of performance including power and confidence interval length comparisons for competing methods. Note that this form of the bootstrap does not entirely escape the problem of estimating the sparsity and its attendant bandwidth issues; it merely advances these issues to an earlier stage of the process by inserting bandwidth choice into the objective function defining the modified median regression estimator. In this vein, it is perhaps also worth noting that other forms of smoothing, like the local L-statistic approach described in Section 5.2, might also be considered. This approach would be closer to that considered by De Angelis, Hall, and Young (1993b) in the univariate setting.

Buchinsky (1995) reports on an extensive Monte Carlo experiment comparing several variants of the bootstrap and concludes that the (x, y) pair, or design matrix version, performs well. A valuable feature of Buchinsky's experiments involves the decoupling of the size of the bootstrap samples, m, and the size of the original sample, n. The option of choosing $m \neq n$ was emphasized already by Bickel and Freedman (1981), but in much of the early bootstrap literature it is presumed that $m = n$. Buchinsky's results suggested that $m < n$ produced more accurate confidence levels than $m = n$ for the residual bootstrap, whereas performance is about the same for the (x, y) bootstrap. This is interesting for at least two reasons. Computationally, $m < n$ has very significant advantages, especially when n is very large. And from a theoretical standpoint, the observation is quite consonant with recent results of Bickel and Sakov (2000), who show that the $m < n$ bootstrap has significant advantages for approximating the behavior of the sample median. In effect, choosing $m < n$ acts as a smoothing device operating on the empirical distribution function. Bickel and Sakov (1999) suggested data-dependent rules for choosing m; theory suggests $m = kn^{2/5}$ and, rather surprisingly, numerical experiments indicate that the constant k can be chosen as small as one. For samples of size 1000, they find empirically optimal m: for the normal distribution, $m = 110$; for Cauchy, $m = 26$; and for logistic, $m = 14$. Compromising on an intermediate value, say $k = 5$, would yield $m \approx 80$ for $n = 1000$ and $m \approx 500$ for $n = 100{,}000$. Subsampling also appears to be an attractive option in inference based on the quantile regression process.

Another approach to refinement of inference for quantile regression is provided by the saddlepoint methods introduced by Daniels (1954). Spady (1991) has explored the saddle-point approach for the median regression estimator when the underlying conditional density of the response is bimodal. The saddlepoint approximation performs extremely well in this setting closely mimicking the multi-modal behavior seen in the Monte-Carlo simulations. Recent work by de Jongh, de Wet, and van Deventer (2003) reenforces this positive impression of saddlepoint methods for quantile regression applications.

3.9.2 Resampling on the Subgradient Condition

The subgradient

$$S_n(b) = -n^{-1/2} \sum_{i=1}^{n} x_i \psi_\tau \left(y_i - x_i^\top b \right)$$

is a pivotal statistic when evaluated at $b = \beta(\tau)$ in the linear quantile regression model; that is, it has a distribution that is, independent of the parameter $\beta(\tau)$. Replacing $\beta(\tau)$ with an estimate yields an asymptotically pivotal quantity. This observation, due to Parzen, Wei, and Ying (1994), has been the source of a number of innovative resampling proposals for quantile regression. Since

$$\psi_\tau \left(y_i - x_i^\top b \right) = \tau - I \left(y_i \le x_i^\top b \right),$$

we can generate independent Bernoulli random variables $\{\xi_i\}$, taking the value 1 with probability τ and 0 with probability $1-\tau$, and compute

$$U_n = -n^{-1/2}\sum_{i=1}^{n} x_i(\tau - \xi_i).$$

For any realization, u_n, of U_n, we can find

$$\beta_n^* = \{b : S_n(b) = u_n\}$$

by solving the augmented quantile regression problem

$$\beta_n^* = \text{argmin}_{b\in\Re} \sum \rho_\tau(y_i - x_i b) + \rho_\tau\left(\zeta - n^{1/2}u_n^\top b/\tau\right),$$

where ζ is chosen to be sufficiently large to insure that it exceeds $n^{1/2}u_n^\top b/\tau$. This ensures that the contribution of the last term to the subgradient is

$$-n^{1/2}u_n = -\sum x_i(\tau - \xi_i)$$

and, thus, that $S_n(b) = u_n$ is satisfied.

To prove that $\sqrt{n}(\beta_n^* - \hat{\beta}_n)$ has the same asymptotic distribution as $\sqrt{n}(\hat{\beta}_n - \beta_0)$, it is first shown that, for $\| b - \beta_0 \| \leq n^{-1/3}$ and $\| b^* - \beta_0 \| < n^{-1/3}$,

$$\begin{aligned}\sup \| S_n(b^*) - S_n(b) - n^{-1/2}\sum x_i \left[\left(F_i\left(x_i^\top b^* - \beta_0\right)\right)\right. \\ \left. - F_i\left(x_i^\top(b-\beta_0)\right)\right] \| \to 0\end{aligned}$$

and then that

$$\begin{aligned} n^{-1/2}\sum x_i \left[F_i\left(x_i^\top(b^*-\beta_0)\right) - F_i\left(x_i^\top(b-\beta_0)\right)\right] \\ = n^{-1}\sum (f_i(0)+o(1))x_i x_i^\top \sqrt{n}(b^*-b).\end{aligned}$$

Now taking $b^* = \beta_n^*(\tau)$ and $b = \hat{\beta}_n(\tau)$ and noting that $S_n(\beta_n^*(\tau)) - S_n(\hat{\beta}_n(\tau)) = U_n$, we have that

$$\sqrt{n}(\beta_n^*(\tau) - \hat{\beta}_n(\tau)) \rightsquigarrow D_1^{-1}U_n,$$

from which the result follows.

The advantages of this resampling strategy have been exploited in recent work of Bilias, Chen, and Ying (2000) for the censored quantile regression model. For this model,

$$Q_{Y_i}(\tau|x_i) = \max\{0, x_i^\top \beta\}.$$

Powell (1986) suggested the estimator

$$\hat{\beta}_n(\tau) = \text{argmin}_{b\in\Re^p}\sum \rho_\tau\left(y_i - \max\left\{0, x_i^\top b\right\}\right),$$

which has very attractive statistical properties. However, as we have seen, computation of the Powell estimator is complicated by the nonconvexity of the objective function and this makes direct application of the $(x,\ y)$ bootstrap problematic. This problem has been investigated in considerable detail by

Fitzenberger (1996). However, a simple modification of the (x, y) bootstrap based on computing

$$\hat{\beta}_n^*(\tau) = \text{argmin}_{b \in \Re^p} \sum \rho_\tau \left(y_i^* - x_i^\top b\right) I(x_i^* \hat{\beta}_n(\tau) > 0)$$

for (x_i^*, y_i^*) drawn with replacement from the empirical distribution of the $\{(x_i, y_i) : i = 1, \ldots, n\}$ succeeds in providing a bootstrap distribution that is correct to first order for $\sqrt{n}(\hat{\beta}_n(\tau) - \beta(\tau))$. Note that $\hat{\beta}_n(\tau)$ is fixed in the indicator function so that computing $\hat{\beta}_n^*(\tau)$ can be accomplished with conventional linear quantile regression algorithms, thus avoiding the nonconvexities introduced by the piecewise linear form of the response function.

A related approach has recently been proposed by He and Hu (2002). They consider generating a Markov chain resampler based on solutions to a marginal coefficient-by-coefficient version of the gradient condition for general M-estimation problems. This approach has been implemented for quantile regression by Kocherginsky, He, and Mu (2004) and seems especially attractive in large problems with high parametric dimension.

3.10 MONTE CARLO COMPARISON OF METHODS

I will conclude this chapter with a report on a small Monte Carlo experiment designed to compare the performance of some of the methods described earlier. I focus primarily on the computationally less demanding Wald and inverted rankscore methods and three of the resampling methods. Preliminary results indicated that the Hall and Sheather bandwidths performed considerably better than the Bofinger choice and so the reported results are restricted to this choice of bandwidth. We also focus exclusively on the problem of confidence intervals for the median regression parameters, partly because this is the most representative problem and also because it restricts the amount of computation and reporting required. Two models will be considered: one is a pure location-shift model the other involves a more complicated location–scale shift. All eight of the methods considered are implemented as options in the function `summary.rq` of the Quantreg package described in Appendix A. They include the following:

- `Riid` – rank test inversion assuming iid errors
- `Rnid` – rank test inversion assuming independent, not identically distributed (nid) errors
- `Wiid` – Wald interval assuming iid errors, with scalar sparsity estimate
- `Wker` – Wald interval assuming nid errors, with Powell's sandwich estimate
- `Wnid` – Wald interval assuming nid errors, with Siddiqui sandwich estimate
- `Bxy` – Bootstrap interval using (x, y) pair method
- `Bpwy` – Bootstrap interval using Parzen–Wei–Ying method
- `Bmcmb` – Bootstrap interval using Markov chain marginal bootstrap method

Table 3.3. *Confidence interval performance: model 1*

	`Riid`	`Rnid`	`Wiid`	`Wker`	`Wnid`	`Bxy`	`Bpwy`	`Bmcmb`
Coverage								
$n = 100$	0.889	0.897	0.871	0.981	0.860	0.919	0.936	0.909
$n = 500$	0.892	0.895	0.880	0.930	0.837	0.900	0.909	0.880
Length								
$n = 100$	0.313	0.322	0.308	0.464	0.303	0.355	0.367	0.337
$n = 500$	0.119	0.120	0.116	0.143	0.120	0.126	0.128	0.122
CPU time								
$n = 100$	0.052	0.206	0.041	0.012	0.039	0.091	0.110	0.128
$n = 500$	0.221	0.466	0.050	0.017	0.104	0.477	0.564	0.471

The Wald and bootstrap intervals are all based on estimating standard deviations; the rank inversion intervals are found by the parametric programming algorithm described in Section 3.5.5.

3.10.1 Model 1: A Location-Shift Model

As our first model, we consider a classical linear regression model in which the covariates have a pure location shift effect:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + u_i. \tag{3.30}$$

Quite arbitrarily, I've taken $p = 3$ and u_is to be iid Student t on three degrees of freedom, and each of the covariates is also independently generated from the $t(3)$ distribution. The coefficients are all set to zero and so the conditional median of the response given the covariates is identically zero. In Table 3.3 we report results for experiments for two different sample sizes, focusing exclusively on the conditional median. In all cases we report results for confidence intervals that are intended to have coverage 0.90. The results are based on 1000 replications of the simulation experiment. The bootstrap results are each based on 100 bootstrap replications. The reported timings are in fractions of a second for each method based on mean timings over the 1000 simulation realizations. The reported coverage and lengths are based on averaging over the 1000 realizations and over the three slope coefficients of the model. Reported confidence interval lengths are median lengths of the 3000 realizations; mean lengths are quite similar except in the case of the `Rnid` method, which has a small number of realizations with infinite length.

All of the methods perform quite well for this baseline model. The two Wald methods designed to adapt to non-iid error models exhibit some size distortion, but the remaining methods have quite accurate size and quite similar mean lengths. The Wald methods have a clear advantage from a computational efficiency viewpoint.

Table 3.4. *Confidence interval performance: model 2 (linear version)*

	Riid	Rnid	Wiid	Wker	Wnid	Bxy	Bpwy	Bmcmb
Coverage								
$n = 100$ b0	0.765	0.889	0.824	0.898	0.731	0.912	0.917	0.950
$n = 100$ b1	0.844	0.901	0.382	0.801	0.838	0.884	0.892	0.891
$n = 500$ b0	0.827	0.891	0.889	0.899	0.792	0.903	0.891	0.943
$n = 500$ b1	0.832	0.898	0.532	0.865	0.857	0.880	0.881	0.880
Length								
$n = 100$ b0	1.625	2.261	1.897	2.400	1.584	2.359	2.384	2.784
$n = 100$ b1	3.444	4.207	1.230	3.717	3.558	4.132	4.175	4.166
$n = 500$ b0	0.696	0.817	0.818	0.879	0.670	0.857	0.849	1.028
$n = 500$ b1	1.448	1.697	0.725	1.642	1.479	1.684	1.659	1.683
CPU time								
$n = 100$	0.039	0.118	0.041	0.012	0.035	0.056	0.073	0.085
$n = 500$	0.103	0.225	0.047	0.015	0.091	0.274	0.359	0.256

3.10.2 Model 2: A Location–Scale-Shift Model

In model 2 we consider a case in which a single covariate has an effect on both the location and scale of the conditional distribution of the response variable,

$$y_i = \beta_0 + \beta_1 x_i + (1 + x_i)^2 u_i. \tag{3.31}$$

In this model we take the u_i as iid standard normal, and the x_is are generated as $\chi^2_3/3$. As in the case of model 1, the design is drawn only once for the experiment. Again we take the coefficients $\beta_0 = \beta_1 = 0$, and so the conditional median of the response is identically zero. However, in contrast to model 1, where other conditional quantile functions of the response are parallel to the conditional median function and thus only the intercept parameter depends on the quantile specified, now quantile functions other than the median are quadratic in the covariate x. We consider two variants of this model for purposes of evaluating the confidence interval methods: one in which the quantile regression model is specified as linear in x at the median, the other in which it is specified as quadratic in x. In this phase of the experiment, it is more interesting to distinguish the performance of the various coefficients of the model, and so they are reported separately rather than averaging over them as in Table 3.3.

Table 3.4 reports the results from model 2, treating the estimated model as linear in variables. Not surprisingly, the methods based on the iid error assumption now perform rather poorly, but the rank inversion intervals based on the local density model perform quite well, as do the the bootstrap methods. Note that in this case the conditional quantile functions of the adjacent $\tau \pm h$ quantiles are assumed to be linear but are actually quadratic in x, and so the performance is better than might be expected.

Table 3.5. *Confidence interval performance: model 2 (quadratic version)*

	Riid	Rnid	Wiid	Wker	Wnid	Bxy	Bpwy	Bmcmb
Coverage								
$n = 100$ b0	0.712	0.850	0.883	0.894	0.825	0.906	0.932	0.935
$n = 100$ b1	0.696	0.852	0.478	0.727	0.824	0.875	0.907	0.820
$n = 100$ b2	0.752	0.846	0.297	0.514	0.776	0.863	0.912	0.802
$n = 500$ b0	0.747	0.878	0.891	0.874	0.840	0.905	0.913	0.923
$n = 500$ b1	0.774	0.898	0.530	0.812	0.868	0.896	0.906	0.859
$n = 500$ b2	0.804	0.895	0.289	0.661	0.853	0.881	0.900	0.842
Length								
$n = 100$ b0	1.790	2.590	2.505	2.394	2.404	2.685	2.892	3.079
$n = 100$ b1	5.872	8.352	3.742	5.824	8.241	8.939	9.775	8.280
$n = 100$ b2	3.280	4.394	1.049	1.725	4.217	4.730	5.202	4.366
$n = 500$ b0	0.852	1.153	1.168	1.087	1.145	1.212	1.234	1.337
$n = 500$ b1	2.901	3.998	1.782	3.154	3.940	4.151	4.229	3.734
$n = 500$ b2	1.702	2.178	0.526	1.379	2.195	2.291	2.353	2.107
CPU time								
$n = 100$	0.046	0.165	0.042	0.012	0.039	0.076	0.095	0.107
$n = 500$	0.178	0.392	0.049	0.016	0.100	0.402	0.493	0.363

Table 3.5 reports the results from model 2, treating the estimated model as quadratic in variables. Now it is of interest to distinguish the three separate coefficients. Again the rank-based intervals assuming nid errors perform quite well, as do the bootstrap procedures. The Wald nid intervals are somewhat overly optimistic. The computational cost of the rank-based methods is roughly comparable for the rank and bootstrap methods; however, larger sample sizes would reveal that the cost of the rank approach grows more rapidly with n than it does for the bootstrap.

3.11 PROBLEMS

Problem 3.1. Suppose $X_1, \dots X_n$ are iid from df F with $f(x) = F'(x) > 0$ on the real line. Generalize (3.5) for a single quantile to obtain result (3.6) on the joint asymptotic distribution of several sample quantiles.

Problem 3.2. Let $X_1, \dots, X_n$ be iid from F and denote the order statistics $X_{(1)}, \dots, X_{(n)}$. Show that the probability that random interval $I_n(r) = [X_{(r)}, X_{(s)}]$ for $1 \le r \le s = n - r + 1$ covers the τth quantile $\xi_\tau = F^{-1}(\tau)$ is

$$\begin{aligned} P(\xi_\tau \in I_n(r)) &\equiv P\left(X_{(r)} \le \xi_\tau \le X_{(s)}\right) \\ &= P\left(X_{(r)} \le \xi_\tau) - P(X_{(s)} < \xi_\tau\right) \\ &\ge C_n(r), \end{aligned}$$

where

$$C_n(r) = \sum_{i=r}^{n-r} \binom{n}{i} \tau^i (1-\tau)^{n-1}.$$

Equality holds in the last line if and only if (iff) F is continuous at ξ_τ.

Problem 3.3. An interesting extension of Problem 3.2 is provided by Guilbaud (1979), who shows that, for intervals of the form

$$I_n(r,t) = \left[\frac{X_{(r)} + X_{(r+t)}}{2}, \frac{X_{(s)} + X_{(s+t)}}{2} \right],$$

where $1 \leq r \leq s = n - r + 1$, and $0 \leq t < s - r$, we have, *for the median,*

$$P(\xi_{1/2} \in I_n(r,t)) \geq \frac{1}{2} C_n(r) + \frac{1}{2} C_n(r+t)$$

for general F, and for continuous, strictly increasing F,

$$P(\xi_{1/2} \in I_n(r,t)) \leq C_{n-t}(r)$$

with equality in the latter expression iff F is symmetric.

Problem 3.4. Show that the interval in Problem 3.1 may be interpreted as an inversion of the following sign test of the hypothesis $H_0 : \xi_\tau = \xi$: let Q_n be the number of observations $\{X_1, \ldots, X_n\}$ less than ξ. Because Q_n is Bin (n, τ), we can choose $\gamma \in (0, 1)$ and c_α such that, under H_0,

$$P(Q_n < c_\alpha)) + \gamma P(Q_n = c_\alpha) = \alpha$$

and reject H_0 when $Q_n < c_\alpha$ or with probability γ when $Q_n = c_\alpha$, for a one-sided alternative. For a two-sided alternative, we choose a corresponding upper critical value as well. Note that for n moderate to large we can approximate the binomial probabilities with their normal approximation (see e.g., Lehmann, 1959, Problem 3.28).

Problem 3.5. To explore a simple version of the asymptotics for quantile regression consider the scalar parameter (regression through the origin model)

$$y_i = x_i \beta + u_i,$$

where the u_is are iid and satisfy the F conditions of Problem 3.1. Formulate conditions on the design sequence $\{x_i\}$ which ensure that

$$\hat{\beta}_n = \text{argmin}_{b \in \mathbb{R}} \sum_{i=1}^{n} |y_i - x_i b|$$

satisfies

$$\sqrt{n}(\hat{\beta}_n - \beta) \rightsquigarrow N(0, \omega^2 Q^{-1})$$

and $Q = \lim n^{-1} \sum x_i^2$.

Problem 3.6 (Bofinger 1975). Show using standard density estimation techniques that the optimal bandwidth for minimum mean-squared error estimation of the sparsity function at τ is

$$h_n = n^{-1/5}\left(\frac{4.5s^2(t)}{(s''(t))^2}\right)^{1/5}.$$

Of course, if we knew $s(t)$ and $s''(t)$ we wouldn't need h_n, but fortunately $s(t)/s''(t)$ is not very sensitive to F. Show that h_n is invariant to location and scale of F, for example. Compare h_n for some typical distributional shapes – say, the Gaussian, Student, and log-normal. Show

$$\frac{s(t)}{s''(t)} = \frac{f^2}{2(f'/f)^2 + [(f'/f)^2 - f''/f]}$$

and, for example, if f is Gaussian, $(f'/f)(F^{-1}(t)) = -\Phi^{-1}(t)$ so that the term in square brackets is 1, and the optimal bandwidth becomes

$$h_n = n^{-1/5}\left(\frac{4.5\phi^4(\Phi^{-1}(t))}{(2\Phi^{-1}(t)^2 + 1)^2}\right)^{1/5}.$$

Plot this bandwidth, as a function of n for several quantiles, and compare the plots across the distributions. Sheather and Maritz (1983) discuss preliminary estimation of s and s'' as a means of estimating a plug-in h_n.

Problem 3.7. Compare the Sheather–Hall bandwidth rule given in the text with the Bofinger bandwidth of the preceding problem.